

Welcome to the SIE Exam Prep Course - 2026 Edition

Start Your Journey in Finance Today

What is the SIE Exam?

- Administered by FINRA
- Tests foundational securities industry knowledge
- Covers investment products, market structure, rules, and regulations

What to Expect in the Exam

- **Format:** 80 multiple-choice questions
 - 75 scored
 - 5 unscored (FINRA pilot questions)
- **Time:** 1 hour and 45 minutes
- **Passing Score:** 70% or higher.

Key Topics Covered in the Exam

The exam is divided into four main sections:

1. Knowledge of Capital Markets (16%)
Market structures, economic factors, offerings, and regulatory agencies.
2. Understanding Products and Their Risks (44%)
Securities products, investment returns, and risk factors.
3. Trading, Customer Accounts, and Prohibited Activities (31%)
Order types, account types, and ethical considerations.
4. Overview of the Regulatory Framework (9%)
FINRA, SEC rules, and key regulatory acts.

How This Course Helps You Pass The SIE

- Breaks down complex securities concepts into clear, easy-to-follow lessons
- Reinforces learning with practice questions after every section
- Includes 2 full-length practice exams that mirror the real SIE
- Builds confidence and knowledge to walk into the exam prepared
- Provides downloadable study resources: slides, cheat sheets, and key exam references

Capital Markets

What Are Capital Markets?

- Platforms for buying and selling long-term financial instruments
- Connect capital seekers (companies, governments) with capital providers (investors)
- Distinction from money markets: long-term debt and equity vs short-term instruments
- Enable price discovery through supply and demand

Why Capital Markets Matter

- Drive economic growth by channeling savings into productive investments
- Enable businesses to raise funds for expansion
- Allow governments to finance infrastructure and public projects
- Provide investors opportunities to grow wealth
- Facilitate efficient allocation of resources

Overview of Market Types

- Four types of markets in the securities industry
- Primary Market: New securities issued
- Secondary Market: Existing securities traded
- Third Market: Listed securities traded OTC
- Fourth Market: Direct institutional trading

The Primary Market

- Where securities are created and sold for the first time
- Issuer receives the proceeds from the sale
- Investment banks act as underwriters
- Examples: IPOs, bond issuances, follow-on offerings
- Governed by Securities Act of 1933

Primary Market Characteristics

- Direct interaction between issuer and investors
- Underwriters help price and distribute securities
- 20-day cooling-off period after SEC filing
- Indications of interest gathered, no sales during cooling-off
- Proceeds used for business operations, expansion, or debt repayment

The Secondary Market

- Trading of existing securities between investors
- Issuer does not receive proceeds
- Provides liquidity for investors
- Determines market prices through trading activity
- Governed by Securities Exchange Act of 1934

Secondary Market Venues

- Exchanges: Physical or electronic auction markets
- NYSE: Uses Designated Market Makers (DMMs/Specialists)
- NASDAQ: Electronic dealer market with multiple market makers
- OTC Markets: Less regulated, includes OTCBB and Pink Sheets
- Each venue has different listing requirements and oversight

The Third Market

- Exchange-listed securities traded over-the-counter
- Occurs outside of the exchange floor
- Used primarily by institutional investors
- Trades are included in exchange volume totals
- Allows for negotiated prices and larger block trades

The Fourth Market

- Direct trading between institutional investors
- Bypasses traditional broker-dealers
- Electronic Communications Networks (ECNs)
- Dark pools: Private exchanges with no public quote display
- Benefits: Anonymity, reduced market impact, lower costs

Market Structure Comparison

- Primary: New issuance, issuer receives proceeds
- Secondary: Existing securities, investors trade with each other
- Third: Listed securities traded OTC, institutional focus
- Fourth: Direct institution-to-institution, ECNs and dark pools
- Key distinction: Who receives the proceeds and where trading occurs

Primary vs Secondary Markets

- Primary: First-time sale, capital raised for issuer
- Secondary: Subsequent trading, liquidity for investors
- Primary governed by Securities Act of 1933
- Secondary governed by Securities Exchange Act of 1934
- Both essential for functioning capital markets

Capital Markets Key Takeaways

- Capital markets connect those who need capital with those who have it
- Four market types: Primary, Secondary, Third, Fourth
- Primary market: Issuer receives proceeds
- Secondary market: Investors trade with each other
- Third and Fourth markets serve institutional investors
- Understanding market structure is foundational for the SIE exam

Capital Markets

Regulatory Entities & Agencies

Why Capital Markets Need Regulation

- Protects investors participating in capital markets
- Maintains fair, orderly, and efficient markets
- Promotes transparency in securities offerings
- Builds confidence for issuers and investors
- Enables capital formation to drive economic growth

The Regulatory Hierarchy

- Federal Government: SEC at the top
- Self-Regulatory Organizations: Operate under SEC supervision
- State Regulators: Parallel authority through Blue Sky laws
- Each level has distinct jurisdiction and powers
- Understanding who regulates what is key for the exam

The SEC: Federal Oversight of Capital Markets

- Created by Securities Exchange Act of 1934
- Federal government agency (not an SRO)
- Mission: Protect investors, maintain fair markets, facilitate capital formation
- Oversees all SROs including FINRA and MSRB
- Ultimate authority over U.S. securities markets

SEC Jurisdiction

- Primary market: Reviews registration statements for new offerings
- Secondary market: Oversees exchanges and trading activity
- Regulates broker-dealers, investment advisers, investment companies
- Enforces federal securities laws
- Has civil authority; criminal cases go to Department of Justice

Self-Regulatory Organizations (SROs)

- Non-governmental organizations with regulatory authority
- Created by industry to regulate its own members
- Operate under SEC supervision and approval
- Write and enforce rules for member firms
- Reduce burden on government regulators

FINRA: Regulating Broker-Dealers

- Financial Industry Regulatory Authority
- Largest SRO for the securities industry
- Regulates broker-dealers and registered representatives
- Oversees firms in primary and secondary markets
- Administers qualification exams including the SIE
- Operates under SEC oversight

MSRB: Municipal Securities Oversight

- Municipal Securities Rulemaking Board
- Writes rules for municipal securities dealers and advisors
- Operates EMMA (Electronic Municipal Market Access)
- Key exam point: MSRB cannot enforce its own rules
- FINRA and SEC enforce for broker-dealers
- Banking regulators enforce for bank dealers

CBOE: Options Market Regulation

- Chicago Board Options Exchange
- Largest U.S. options exchange
- SRO with authority over options trading
- Regulates options market participants
- Operates under SEC oversight

SIPC: Protecting Brokerage Customers

- Securities Investor Protection Corporation
- Protects customers if a brokerage firm fails
- Not a government agency (non-profit funded by members)
- Not an SRO (does not write or enforce rules)
- Distinct from FDIC which insures bank deposits

Federal Reserve Board

- Central bank of the United States
- Controls monetary policy and interest rates
- Sets credit rules for securities purchases (Regulation T)
- Impacts capital markets through policy decisions
- Not primarily a securities regulator

State Regulators and Blue Sky Laws

- Each state has its own securities regulator
- Blue Sky laws: State-level securities regulations
- Regulate offerings and professionals within the state
- Work alongside federal regulation
- NASAA coordinates state regulators (not a regulator itself)

Other Federal Agencies

- Treasury Department: Economic and tax policy
- FDIC: Insures bank deposits (not brokerage accounts)
- Department of Justice: Criminal prosecution of securities fraud
- OCC: Regulates national banks
- Support but do not directly oversee capital markets

Who Regulates What: Quick Reference

- SEC: Overall securities markets, federal oversight
- FINRA: Broker-dealers and registered representatives
- MSRB: Municipal securities rules (but no enforcement)
- CBOE: Options exchanges and trading
- States: Securities within their borders
- Federal Reserve: Credit and margin requirements

Key Takeaways: Regulatory Entities

- SEC is the federal government agency overseeing capital markets
- SROs regulate members under SEC supervision
- FINRA is the largest SRO for broker-dealers
- MSRB writes rules but cannot enforce them
- SIPC protects investors but is not a regulator
- States have parallel authority through Blue Sky laws

SIE Exam Tips: Regulatory Entities

- SEC is government; FINRA, SIPC are not
- MSRB enforcement limitation is heavily tested
- Know each agency's jurisdiction and scope
- Understand the hierarchy: SEC → SROs → Members
- Blue Sky laws = state securities regulation

Capital Markets Players and Roles

Capital Market Players – Introduction

- Capital markets connect those who need capital (**issuers**) with those who provide it (**investors**)
- Intermediaries facilitate transactions between issuers and investors
- Every participant has a distinct role, registration, and regulatory oversight
- Understanding "who does what" is foundational for SIE success

Buy-Side vs. Sell-Side: The Market's Two Halves

- **Sell-Side:** Creates and distributes securities (investment banks, broker-dealers, underwriters). Earn fees/commissions
- **Buy-Side:** Purchases securities for investment (mutual funds, pension funds, hedge funds, retail investors). seeks investment returns
- A single firm can have both buy-side and sell-side operations

Types of Investors

- **Retail Investors:** Individuals investing personal funds; protected by Reg BI
- **Institutional:** Mutual funds, pension funds, insurance companies, endowments
- **Accredited Investors:** Meet income (\$200K/\$300K joint) or net worth (\$1M excl. primary residence) thresholds
- **Qualified Institutional Buyers (QIBs):** Institutions with \$100M+ in securities; eligible for Rule 144A offerings

Broker vs. Dealer: Agent vs. Principal

- **Broker (Agent):** Executes trades on behalf of customers; earns commission; does NOT own securities
- **Dealer (Principal):** Trades from own inventory; earns markup/markdown; takes ownership risk
- Most firms are registered as Broker-Dealers (can act in either capacity)
- Firm must disclose capacity on trade confirmations

Types of Broker-Dealers

- **Introducing Broker-Dealer:** Accepts customer orders but does NOT hold custody; clears through another firm
- **Clearing Firm:** Holds customer accounts, settles trades, maintains custody
- **Prime Broker:** Provides consolidated services to hedge funds (clearing, lending, custody, reporting)
- Introducing firms must have clearing agreements with clearing firm

Market Makers: Ensuring Liquidity

- Market makers quote continuous bid (buy) and ask (sell) prices
- Earn profit from the bid-ask spread
- Required to maintain orderly markets by standing ready to buy/sell
- NASDAQ: Multiple competing market makers per security
- NYSE: Designated Market Makers (DMMs) assigned to specific securities

Underwriters & Syndicate Members

- **Underwriters:** Investment banks that help issuers bring securities to market
- **Firm Commitment:** Underwriter buys entire issue, bears full risk if unsold
- **Best Efforts:** Underwriter sells what it can; unsold shares returned to issuer
- **All-or-None:** Offering cancelled if entire issue not sold
- **Syndicate:** Group of underwriters sharing distribution and risk
- **Municipal Advisors:** Advise issuers on municipal offerings; MSRB-regulated

Custodians, Transfer Agents & Other Players

- **Custodians:** Hold and safeguard securities on behalf of investors (banks, clearing firms)
- **Transfer Agents:** Maintain records of security ownership, issue/cancel certificates, process transfers
- **Trustees:** Oversee bond indentures, protect bondholder interests
- **Paying Agents:** Distribute interest and dividend payments
- **These roles are distinct:** Custodians hold; Transfer agents record

Issuers: Who Raises Capital

- **Corporations:** Issue stocks and bonds for expansion, operations, acquisitions
- **Governments:** U.S. Treasury issues bills, notes, bonds
- **Municipalities:** Issue municipal bonds for public projects
- **Agencies:** GSEs (Fannie Mae, Freddie Mac) issue agency securities
- **Issuers are NOT intermediaries**—they are capital seekers

Capital Market Participants: Quick Reference

- **Broker:** Agent; executes customer orders → Commission
- **Dealer:** Principal; trades own inventory → Markup/Markdown
- **Market Maker:** Provides liquidity; quotes bid/ask → Spread
- **Underwriter:** Distributes new issues → Underwriting fee
- **Custodian:** Safeguards securities → Custody fees
- **Transfer Agent:** Records ownership → Service fees

Security Offerings

Public Offering (IPO)_s

- A company sell shares to the public, typically during an Initial Public Offering. Allows access to a wide pool of investors for significant capital-raising.
- Regulated to ensure transparency and protect investors.

Benefits:

- Access to substantial capital for growth.
- Increased visibility and market credibility.
- Shares are easily tradable on public exchanges, ensuring liquidity.

Challenges:

- High regulatory compliance with SEC requirements.
- Increased public scrutiny and mandatory disclosure obligations.

Private Offerings

- Securities (Shares or Bonds) are sold directly to select investors (e.g., institutional investors, wealthy individuals).
- Less regulated and faster than public offerings.
- Not offered on public exchanges, avoiding broader market involvement.

Benefits:

- Faster process with reduced regulatory burdens.
- Flexibility in structuring terms and deals with investors.

Challenges:

- Limited pool of investors reduces fundraising potential.
- Securities are less liquid, making resale challenging for investors.

How an IPO works?

1. **Preparation:**

Collaboration with investment banks (underwriters) to determine share price and structure.

2. **Regulatory Filing:**

Filing detailed financial and operational data with the SEC.

3. **Public Listing:**

Shares listed on a stock exchange (e.g., NYSE, Nasdaq).
Trading begins.

Follow-On and Secondary Offerings

Post-IPO mechanisms to raise funds or allow shareholders to sell shares, expand capital access or increase share liquidity.

Follow-On Offering

- Company issues additional shares to raise capital to fund new projects, acquisitions, or pay down debt.
- Risk: Dilution of existing shares, potentially reducing their value.

Secondary Offering

- Existing shareholders (e.g., founders, early investors) sell their shares.
- No new funds raised for the company. Increases liquidity of shares for market participants.

Economic Factors & the Federal Reserve

Learning Objectives:

- Distinguish monetary policy from fiscal policy
- Explain the Fed's tools and their market impact
- Identify leading, lagging, and coincident indicators
- Understand business cycle stages

Monetary vs. Fiscal Policy

- Monetary Policy: Controlled by Federal Reserve
 - Tools: Interest rates, money supply
- Fiscal Policy: Controlled by Congress/President
 - Tools: Taxes, government spending
- SIE Exam Tip: Fed = monetary policy. Congress = fiscal policy. Know the difference.

The Federal Reserve's Role

- Central bank of the United States
- Controls money supply and interest rates
- Goals: Maximum employment, stable prices, moderate long-term rates
- Sets credit rules for securities purchases (Regulation T)
- SIE Exam Tip: Fed manages monetary policy; Reg T sets 50% initial margin requirement.

Fed's Three Main Tools

- Open Market Operations: Buy/sell Treasury securities
- Discount Rate: Rate Fed charges banks for loans
- Reserve Requirements: % banks must hold in reserve
- SIE Exam Tip: Open market operations is the Fed's most frequently used tool.

Open Market Operations

- Fed BUYS securities: Injects money, increases supply (Expansionary)
- Fed SELLS securities: Removes money, decreases supply (Contractionary)
- Expansionary = stimulates economy
- Contractionary = slows economy
- SIE Exam Tip: Fed buys = expansionary = more money. Fed sells = contractionary = less money.

Key Interest Rates

- Federal Funds Rate: Rate banks charge each other overnight
- Discount Rate: Rate Fed charges banks (higher than fed funds)
- Prime Rate: Rate banks charge best customers (tracks fed funds)
- Relationship: Discount Rate > Prime Rate > Fed Funds Rate
- SIE Exam Tip: Fed funds rate is the key benchmark short-term rate.

Interest Rates and Markets

- Rates UP → Bond prices DOWN
- Rates UP → Stock prices generally DOWN (higher borrowing costs)
- Rates DOWN → Stimulates borrowing, spending, investment
- SIE Exam Tip: Interest rates and bond prices move inversely—always.

Business Cycle Stages

- Peak → Contraction → Trough → Expansion → Peak
- Expansion: Economy growing, rising employment and inflation
- Peak: Economy at maximum, high employment and inflation
- Contraction: Economy shrinking, falling employment and inflation
- Trough: Economy at minimum, low employment and inflation
- SIE Exam Tip: Know the four stages and their characteristics.

Economic Indicators

- Leading: Predicts future (building permits, stock prices, money supply)
- Lagging: Confirms past (unemployment rate, CPI, prime rate)
- Coincident: Shows current (GDP, industrial production, personal income)
- SIE Exam Tip: Leading = predicts. Lagging = confirms. Coincident = current. Memorize examples.

Cyclical vs. Defensive Stocks

- Cyclical: Performance tied to economic cycle
- Examples: Airlines, autos, hotels, luxury goods
- Defensive (Counter-cyclical): Stable regardless of economy
- Examples: Utilities, healthcare, consumer staples
- SIE Exam Tip: Cyclical = discretionary spending. Defensive = necessities.

Inflation and Deflation

- Inflation: Rising prices, decreasing purchasing power
- Erodes fixed-income returns
- TIPS protect against inflation
- Deflation: Falling prices, increases real value of debt
- SIE Exam Tip: Inflation hurts bondholders (fixed payments buy less).

International Factors

- GDP: Domestic output (goods/services produced within country)
- Balance of Payments: Trade surplus (exports > imports) vs. deficit
- Strong dollar → Imports cheaper, exports more expensive
- Weak dollar → Exports cheaper, imports more expensive
- SIE Exam Tip: Strong dollar hurts exporters (foreign buyers pay more).

Key Takeaways

- Fed controls monetary policy (money supply, rates)
- Fed buys = expansionary; Fed sells = contractionary
- Four business cycle stages: expansion, peak, contraction, trough
- Leading indicators predict; lagging confirm; coincident show current
- Cyclical stocks follow economy; defensive stocks remain stable
- Inflation erodes purchasing power of fixed income

Rules and Regulations Governing Capital Markets

Securities Act of 1933

- **Purpose: "Truth in Securities" / Primary market disclosure**
Requires full disclosure of material information for new securities offerings.
- **Section 7: Registration statement requirements**
Specifies what information issuers must file with the SEC before offering securities.
- **Section 10: Prospectus requirements**
Investors must receive a prospectus with key information before or at purchase.
- **Section 23: Unlawful representations (no SEC approval claims)**
SEC reviews for completeness only—claiming SEC "approved" a security is illegal.
- **20-day cooling off period**
Mandatory waiting period after filing; no sales allowed, but indications of interest permitted.

Securities Exchange Act of 1934

- **Secondary market regulation, created SEC**
Regulates trading of securities after initial issuance and established the Securities and Exchange Commission.
- **Section 10(b) / Rule 10b-5: Anti-fraud provisions**
Prohibits fraud, manipulation, and deceptive practices in connection with buying or selling securities.
- **Section 12: Exchange registration requirements**
Securities traded on national exchanges must register with the SEC and meet ongoing reporting standards.
- **Section 14: Proxy solicitation rules**
Requires disclosure of material information when soliciting shareholder votes on corporate matters.
- **Section 15: Broker-dealer registration**
All broker-dealers must register with the SEC and comply with investor protection rules.
- **Section 15A: Self-regulatory organizations**
Authorizes SROs like FINRA to create and enforce rules governing member firms.

Regulation D - Private Placement

- **Purpose: Exemption from full registration** Allows companies to raise capital without costly SEC registration by selling to limited or sophisticated investors.
- **Rule 504: Up to \$10M, no specific disclosure required** Small offerings exempt from registration; state securities laws (Blue Sky) still apply.
- **Rule 506(b): Unlimited amount, up to 35 non-accredited, no general solicitation** Most common exemption; issuer must have pre-existing relationship with investors and provide disclosure to non-accredited.
- **Rule 506(c): Unlimited amount, accredited only, general solicitation permitted** Allows public advertising but issuer must take reasonable steps to verify accredited status.
- **Accredited Investor definition (Rule 215): \$1M net worth or \$200K/\$300K income** Individuals meeting wealth/income thresholds are deemed sophisticated enough to evaluate investment risks.
- **Restricted securities – holding period, resale limitations** Securities purchased in private placements cannot be freely resold; typically held 6-12 months before resale under Rule 144.
- **Form D filing requirement** Issuers must file Form D with SEC within 15 days of first sale to claim the exemption.

Other Exemptions from Registration

- **Rule 144: Resale of restricted/control securities (6-month/1-year holding)**
Allows holders to sell restricted or control securities publicly after meeting holding period and volume limitations.
- **Rule 144A: Resales to Qualified Institutional Buyers (QIBs)**
Permits resale of unregistered securities to large institutions with at least \$100M in assets under management.
- **Rule 147/147A: Intrastate offerings**
Exempts offerings made entirely within one state to residents of that state; issuer must be based there.
- **Regulation A/A+: Mini-registration (\$20M/\$75M tiers)**
Simplified registration allowing smaller companies to raise capital with reduced disclosure requirements.
- **Regulation S:**
Offshore offerings Exempts offerings made outside the U.S. to non-U.S. persons from SEC registration.
- **Regulation Crowdfunding**
Allows companies to raise up to \$5M annually from the general public through SEC-registered funding portals.

Securities Exchange Act of 1934 – Key Rules

- **Rule 15l-1: Regulation Best Interest (Reg BI)**
Broker-dealers must act in retail customers' best interest when making recommendations, without placing their own interests ahead.
- **Rules 17a-3/17a-4:**
Books and records requirements Specifies what records firms must create (17a-3) and how long they must be retained (17a-4: 3-6 years).
- **Section 20A/21A:**
Insider trading liability and penalties Creates civil liability to contemporaneous traders and allows penalties up to 3x profits gained or losses avoided.
- **Rule 10b5-1:** Trading plans (affirmative defense) Pre-established written trading plans created when not in possession of MNPI provide defense against insider trading claims.
- **Rule 10b5-2:** Duties of trust (misappropriation) Trading on confidential information obtained through a relationship of trust violates securities laws even without corporate insider status.

Investment Company Act of 1940

- Regulate pooled investment vehicles Protects investors in mutual funds, ETFs, and other investment companies through disclosure and governance requirements.
- **Types:** Open-end, closed-end, UITs, ETFs Each structure has different share issuance, redemption, and trading characteristics.
- **Section 3(a):** Definition of investment company Defines entities primarily engaged in investing, reinvesting, or trading in securities.
- **Section 4/5:** Classification and subclassification Categorizes investment companies by structure (face-amount, UIT, management) and type (open/closed, diversified/non-diversified).
- **Rule 12b-1:** Distribution fees Allows funds to use shareholder assets to pay for marketing and distribution expenses, capped at 1% annually.
- **Diversified vs. non-diversified requirements** Diversified funds must limit holdings to 5% of assets per issuer for at least 75% of the portfolio.
- **Board composition (40% independent directors)** At least 40% of board members must be independent to protect shareholder interests from management conflicts.

Investment Advisers Act of 1940

- Definition of investment adviser (3-prong test) Anyone who provides advice about securities, for compensation, as part of a regular business.
- Fiduciary duty to clients IAs must act in clients' best interests with duties of care and loyalty, disclosing all material conflicts.
- Registration thresholds: SEC (>\$110M) vs. State (<\$100M) Assets under management determine whether an IA registers federally or with state regulators.
- Form ADV disclosure requirements Two-part disclosure document detailing firm's services, fees, disciplinary history, and conflicts of interest.
- Exclusions: Banks, lawyers, accountants, publishers, B/Ds Certain professionals are excluded if investment advice is incidental to their primary business.
- IA vs. BD distinction (fee-based vs. commission-based) IAs charge fees and owe fiduciary duty; BDs earn commissions and are subject to Reg BI standard.

MSRB Rules – Municipal Securities

- MSRB role: Writes rules, does not enforce (SEC/FINRA enforce) Municipal Securities Rulemaking Board creates regulations but relies on other regulators for enforcement.
- Rule G-11: Primary offering practices Governs allocation and distribution of new municipal securities to ensure fairness and transparency.
- Rule G-17: Fair dealing Dealers must deal fairly with all persons and not engage in deceptive or dishonest practices.
- Rule G-18: Best execution Requires reasonable diligence to obtain the most favorable price for customer transactions.
- Rule G-30: Pricing and commissions All prices and commissions charged to customers must be fair and reasonable.
- Rule G-32: Official statement delivery Dealers must deliver the official statement to customers no later than settlement of the transaction.

SIPA and SIPC

- Securities Investor Protection Act of 1970 Created SIPC to protect customers when broker-dealers fail financially or become insolvent.
- SIPC coverage: \$500,000 total / \$250,000 cash limit Maximum protection per customer account; cash portion cannot exceed \$250,000 of the total.
- What's covered: Securities, cash for securities purchases Protects stocks, bonds, and cash held in brokerage accounts when the firm fails.
- What's NOT covered: Market losses, commodities, futures, annuities SIPC does not insure against investment losses or protect non-securities products.
- SIPC vs. FDIC distinction SIPC covers brokerage account assets from firm failure; FDIC covers bank deposits from bank failure.
- FINRA Rule 2266: SIPC disclosure requirement Broker-dealers must provide customers with information about SIPC coverage at account opening

Introduction to Equity Securities

Learning Objectives:

- Define equity and explain what ownership in a corporation means
- Distinguish equity from debt at a high level
- Identify the two main types of stock

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What Is an Equity Security?

- Represents ownership (a "share") in a corporation
- Stockholders are part-owners of the company
- Value tied to company performance
- Contrasts with debt (lending money)

SIE Exam Tip: Equity = ownership; Debt = lending. Questions often test this distinction.

Why Companies Issue Stock

- Raise capital without taking on debt
- No obligation to repay principal
- Dilutes existing ownership
- Access to public markets (IPO)

Example: Startup raises \$50M by selling 25% of shares to investors.

Two Main Types of Stock

- Common stock: Voting rights, variable dividends, growth potential
- Preferred stock: Fixed dividends, priority over common, limited/no voting
- Both represent equity ownership

SIE Exam Tip: "Preferred" refers to dividend and liquidation priority—not superiority in all ways.

Equity vs. Debt Summary

SIE Exam Tip: Bondholders get paid before stockholders in bankruptcy—always.

Key Takeaways

- Equity = ownership stake in a company
- Common and preferred are the two main types
- Stockholders bear more risk but have unlimited upside
- Bondholders have priority but capped returns

Common Stock: Ownership, Voting & Risk

Learning Objectives:

- Describe the features and rights of common stockholders
- Explain voting rights and proxy voting
- Identify the risks and position in liquidation

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What Is Common Stock?

- Most basic form of equity ownership
- Residual claim on assets and earnings
- Potential for capital appreciation
- Dividends not guaranteed

SIE Exam Tip: "Residual" means common shareholders get what's left after everyone else is paid.

Voting Rights

- One share = one vote (typically)
- Elect board of directors
- Approve major corporate actions (mergers, charter amendments)
- Statutory voting: Votes spread across all candidates
- Cumulative voting: Can concentrate votes on one candidate

SIE Exam Tip: Cumulative voting benefits minority shareholders—this is testable.

Proxy Voting

- Shareholders can vote in person or by proxy
- Proxy = authorization for someone else to vote your shares
- Proxy statement contains meeting agenda, proposals, board info
- Must be delivered before annual meeting

Example: You own 100 shares but can't attend—you sign a proxy card authorizing management to vote for you.

SIE Exam Tip: Proxy solicitation is regulated by SEC.

Dividends on Common Stock

- Declared by board of directors—not guaranteed
- Paid from profits (retained earnings)
- Variable based on company performance
- Can be suspended during downturns

SIE Exam Tip: Common dividends are never "owed"—they're discretionary.

Capital Appreciation

- Stock price can rise as company grows
- Unlimited upside potential
- Also unlimited downside (can go to zero)
- Total return = dividends + price change

Liquidation Priority

- Common stockholders are last in line
- Order: Secured creditors → Unsecured creditors → Preferred stock → Common stock
- May receive nothing in bankruptcy

SIE Exam Tip: Expect questions on liquidation order—common is always last.

Risks of Common Stock

- Market risk: Price fluctuates with market
- Business risk: Company-specific problems
- Dividend risk: Payments can be cut or eliminated
- Liquidity risk: Some stocks hard to sell quickly

Who Should Own Common Stock?

- Investors seeking growth and capital appreciation
- Those with longer time horizons
- Willing to accept volatility
- Want voting influence in company

SIE Exam Tip: Suitability questions tie risk tolerance to security type.

Key Takeaways

- Common stock = ownership + voting + variable dividends
- Last in line during liquidation
- Higher risk, higher potential reward
- Cumulative voting helps minority shareholders

Preferred Stock: Types & Trade-offs

Learning Objectives:

- Explain preferred stock features and how it differs from common
- Distinguish cumulative, callable, convertible, and participating preferred
- Understand the trade-offs preferred stockholders accept

What Is Preferred Stock?

- Hybrid security: features of both equity and debt
- Fixed dividend (stated rate)
- Priority over common for dividends and liquidation
- Generally no voting rights

SIE Exam Tip: Preferred is "equity" for classification - not debt - despite fixed payments.

Preferred vs. Common Comparison

Feature	Common	Preferred
Dividends	Variable	Fixed
Voting	Yes	Usually no
Liquidation	Last	Before common
Growth potential	High	Limited

Cumulative Preferred

- Unpaid dividends accumulate as "dividends in arrears"
- All arrears must be paid before common receives any dividend
- Protects investors during hard times

Example: Company skips 2 years of \$5 dividends. Cumulative preferred holders owed \$10/share before common gets anything.

SIE Exam Tip: "Cumulative" = missed dividends stack up. Heavily tested.

Non-Cumulative Preferred

- Missed dividends are forfeited—gone forever
- Higher risk than cumulative
- May offer slightly higher stated dividend rate

SIE Exam Tip: Non-cumulative preferred holders lose skipped dividends permanently.

Callable Preferred

- Issuer can "call" (redeem) shares at a set price
- Typically called when interest rates fall
- Issuer refinances at lower cost
- Call risk to investor: reinvestment at lower rates

Example: 6% preferred called when rates drop to 4%—investor must reinvest at lower yield.

SIE Exam Tip: Callable benefits the issuer, not the investor.

Convertible Preferred

- Can be converted into common stock at a set ratio
- Offers upside if common stock rises
- Conversion is at investor's option
- Typically pays lower dividend than non-convertible

Example: Convertible at 4:1 ratio. If common rises from \$20 to \$40, converting becomes attractive.

SIE Exam Tip: Convertible gives upside potential but trades off current income.

Participating Preferred

- Receives fixed dividend PLUS may share in extra profits
- Rare; offers best of both worlds
- "Participates" alongside common if company does very well

SIE Exam Tip: Participating = gets fixed dividend AND potential extra.
Uncommon but testable.

Adjustable-Rate Preferred

- Dividend rate adjusts periodically (tied to benchmark rate)
- Reduces interest rate risk
- Price more stable than fixed-rate preferred

Who Should Own Preferred Stock?

- Income-focused investors seeking stability
- Those willing to sacrifice growth for predictable payments
- Lower risk tolerance than common stockholders
- Often attractive to corporate investors (dividend exclusion)

SIE Exam Tip: Corporations get tax break on dividends received—makes preferred attractive to them.

Key Takeaways

- Preferred = fixed dividends + priority + limited voting
- Cumulative: arrears must be paid first
- Callable: issuer can redeem (call risk)
- Convertible: can become common (upside potential)
- Participating: shares extra profits (rare)

Dividends: Types, Dates & Taxation

Learning Objectives:

- Distinguish cash, stock, and property dividends
- Master the four key dividend dates and their sequence
- Understand basic tax treatment of dividends

What Is a Dividend?

- Distribution of company profits to shareholders
- Declared by board of directors
- Typically paid quarterly (for common)
- Not an obligation—discretionary for common

Types of Dividends

- Cash dividend: Most common; paid in dollars per share
- Stock dividend: Additional shares instead of cash (e.g., 10% stock dividend = 10 new shares per 100 owned)
- Property dividend: Rare; paid in assets or subsidiary stock

SIE Exam Tip: Stock dividends increase shares but don't change total value (no taxable event until sold).

The Four Key Dividend Dates

Date	What Happens
Declaration date	Board announces dividend
Ex-dividend date	First day stock trades WITHOUT dividend
Record date	Must be shareholder "of record" to receive
Payable date	Dividend is actually paid

Memory trick: "DERP" = Declaration, Ex-div, Record, Payable

Ex-Dividend Date Deep Dive

- Set by FINRA: 1 business day before record date
- Buy BEFORE ex-date → receive dividend
- Buy ON or AFTER ex-date → do NOT receive dividend
- Stock price typically drops by dividend amount on ex-date

Example: Ex-date is Monday. You must own by Friday's close to get the dividend.

SIE Exam Tip: Ex-dividend date questions are VERY common. "Ex" = excluded from dividend.

Ex-Date Visual Timeline

Declaration → Ex-Dividend → Record → Payable
(Day 1) (Day 10) (Day 11) (Day 25)

- Buy Day 9 = get dividend ✓
- Buy Day 10 = NO dividend ×

Record Date

- Date company checks shareholder list
- Must be "holder of record" on this date
- Settlement takes 1 business day (T+1)
- That's why ex-date is 1 day before record date

SIE Exam Tip: Record date determines WHO gets paid; ex-date determines the cutoff for buying.

Stock Dividends vs. Stock Splits

- Stock dividend: Stated as percentage (e.g., 5%)
- Stock split: Stated as ratio (e.g., 2-for-1)
- Both increase shares, decrease price proportionally
- Neither creates immediate taxable income

Example: 100 shares at \$50 → 5% stock dividend → 105 shares at ~\$47.62

Dividend Taxation Basics

- Qualified dividends: Taxed at lower capital gains rates (must meet holding period)
- Non-qualified (ordinary) dividends: Taxed as ordinary income
- Holding period: 60+ days in 121-day window around ex-date

SIE Exam Tip: SIE tests the distinction exists—not specific rates.

Dividend Reinvestment Plans (DRIPs)

- Automatically reinvest dividends to buy more shares
- Often no commission
- Still taxable even though not received in cash

Key Takeaways

- Three types: cash, stock, property
- Four dates: Declaration, Ex-dividend, Record, Payable
- Ex-date = cutoff for receiving dividend (1 day before record)

Qualified dividends taxed favorably if holding period met

Corporate Actions: Splits, Offerings & Buybacks

Learning Objectives:

- Explain stock splits and reverse splits and their effects
- Describe secondary offerings and share buybacks
- Understand tender offers at a high level

What Is a Stock Split?

- Increases number of shares outstanding
- Decreases price per share proportionally
- Total market value unchanged
- Makes shares more accessible to retail investors

Example: 2-for-1 split: 100 shares at \$80 → 200 shares at \$40

Stock Split Math

- Forward split (2:1, 3:1, etc.): More shares, lower price
- Shares multiply, price divides
- No change in total value or ownership percentage

SIE Exam Tip: Splits don't create or destroy value—just repackage it.

Reverse Stock Split

- Decreases number of shares outstanding
- Increases price per share proportionally
- Often done to meet exchange listing requirements (min price)
- Can signal financial distress

Example: 1-for-10 reverse split: 1,000 shares at \$0.50 → 100 shares at \$5.00

SIE Exam Tip: Reverse splits are often a warning sign—used to avoid delisting.

Effect on Options & Other Securities

- Stock splits adjust options contracts
- Strike prices adjusted, contract quantities adjusted
- Investor's economic position unchanged

Secondary vs Follow-On Offerings

- Follow-on offering: Company issues NEW shares (dilutive)
- Secondary offering: Existing shareholders sell THEIR shares (non-dilutive to company)
- Both increase public float

SIE Exam Tip: Follow-on = company raises capital (dilution); Secondary = insiders sell (no new capital).

Share Buybacks (Repurchases)

- Company buys its own shares on open market
- Reduces shares outstanding
- Increases EPS (fewer shares, same earnings)
- Returns cash to shareholders indirectly

SIE Exam Tip: Buybacks are the opposite of dilution—they concentrate ownership.

Tender Offers (High-Level)

- Public offer to buy shares directly from shareholders
- Usually at premium to market price
- Often used in acquisitions
- Shareholders can accept or reject

Example: Company A offers \$50/share to Company B shareholders when stock trades at \$40.

Spin-Offs

- Parent company distributes shares of subsidiary
- Creates separate publicly traded company
- Existing shareholders receive shares in new company
- Similar to property dividend

Key Takeaways

- Stock split: more shares, lower price, same value
- Reverse split: fewer shares, higher price, often a warning
- Follow-on = dilution; Secondary = insiders selling
- Buybacks reduce shares, increase EPS
- Tender offers = premium bid direct to shareholders

Warrants and Preemptive Rights

Learning Objectives:

- Define preemptive rights and explain their anti-dilution purpose
- Distinguish warrants from rights by duration, pricing, and recipients
- Understand when each is issued and how investors can use them

Two Ways to Buy Future Shares

- Both warrants and rights give holders the option to purchase stock
- Neither represents current ownership—they are contracts for future purchase
- Key differences: WHO receives them, HOW LONG they last, WHAT PRICE
- Understanding the distinction is heavily tested on the SIE

SIE Exam Tip: Both give the RIGHT (not obligation) to buy shares—like options

What Are Preemptive Rights?

- Right of existing shareholders to maintain proportional ownership
- Protects against dilution when company issues new shares
- Also called "subscription rights"—not all companies offer them
- Short-term duration: typically 30–60 days to exercise
- Exercise price set BELOW current market price (discount)

SIE Exam Tip: Rights are SHORT-term and priced BELOW market

How Rights Offerings Work

- Company announces new share issuance to raise capital
- Existing shareholders receive rights (typically 1 right per share owned)
- Rights allow purchase at subscription price (below market)
- Shareholders can: Exercise (buy shares), Sell (trade rights), or Let expire

SIE Exam Tip: Rights can be exercised, sold on exchange, or allowed to expire

Rights Offering Example

- You own 100 shares of XYZ Corporation
- Current market price: \$25 per share
- Rights offering: subscription price of \$20 (below market)
- Terms: 10 rights required to buy 1 new share → You can buy 10 shares

Example: Buy at \$20, worth \$25 = \$5 instant value per share

SIE Exam Tip: Rights have intrinsic value = Market Price – Subscription Price

What Are Warrants?

- Long-term options to purchase stock at a fixed price
- Issued by the company itself (not OCC like listed options)
- Usually attached to bonds or preferred stock as "sweetener"
- Exercise price typically ABOVE current market price at issuance
- Duration: Years (can be up to 10+ years)

SIE Exam Tip: Warrants are LONG-term with strike prices ABOVE market at issuance

Warrant Example

- Current stock price: \$20 | Warrant exercise price: \$25 | Duration: 5 years
- Year 1: Stock at \$22 → Warrant out-of-the-money (no intrinsic value)
- Year 3: Stock rises to \$35 → Warrant in-the-money
- Intrinsic value = $\$35 - \$25 = \$10$ per share profit potential

Example: If stock never exceeds \$25, warrant expires worthless

SIE Exam Tip: Warrants have value only if stock rises above exercise price

Rights vs. Warrants Comparison

- Duration: Rights = Short (30-60 days) | Warrants = Long (years)
- Exercise Price: Rights = Below market | Warrants = Above market
- Issued To: Rights = Existing shareholders | Warrants = Bond/preferred buyers
- Purpose: Rights = Anti-dilution | Warrants = Incentive/sweetener

SIE Exam Tip: This comparison is HEAVILY tested. Memorize it!

Common Exam Traps

- Trap 1: Confusing duration—Rights = SHORT, Warrants = LONG
- Trap 2: Confusing exercise price—Rights = BELOW, Warrants = ABOVE market
- Trap 3: Confusing recipients—Rights to shareholders, Warrants to bondholders
- Trap 4: Warrants $\neq$ Listed Options—Warrants are company-issued and dilutive

SIE Exam Tip: When exercised, both create NEW shares (dilution occurs)

Warrants and Rights - Key Takeaways

- Rights = Short-term (30-60 days), below market, for existing shareholders
- Warrants = Long-term (years), above market, attached as sweetener
- Both give the RIGHT (not obligation) to buy shares
- Both are DILUTIVE when exercised (company issues new shares)
- Rights protect ownership; Warrants add value to bond offerings

What Is an ADR?

- American Depositary Receipt
- U.S.-traded security representing shares of a foreign company
- Priced in USD, pays dividends in USD
- Eliminates need to trade on foreign exchanges

Example: Toyota ADR trades on NYSE—you own economic interest in Toyota without buying on Tokyo exchange.

How ADRs Work

- U.S. bank holds foreign shares in custody
- Issues receipts (ADRs) to U.S. investors
- Each ADR may represent 1 share, multiple shares, or a fraction
- Dividends converted to USD (minus fees)

Sponsored vs. Unsponsored ADRs

Feature	Sponsored	Unsponsored
Issued by	One depositary bank with company cooperation	Multiple banks without company involvement
SEC registration	Usually registered	Often exempt
Disclosure	Full financial reporting	Limited
Exchange listing	Often listed (NYSE, Nasdaq)	OTC only

ADR Risks

- Currency risk: USD/foreign currency fluctuations
- Political risk: Foreign government instability
- Information risk: Less disclosure for unsponsored
- Liquidity risk: May trade less actively

SIE Exam Tip: Currency risk is the unique ADR risk most often tested.

ADR Levels

- Level I: OTC trading, minimal SEC requirements
- Level II: Listed on exchange, full SEC registration
- Level III: Can raise new capital in U.S.

SIE Exam Tip: Higher level = more regulatory requirements and transparency.

Restricted Stock, Control Stock & Rule 144

What Is Restricted Stock?

- Stock acquired in private placement or as compensation
- Cannot be sold publicly without registration or exemption
- Must hold for minimum period before selling
- Legend on certificate indicates restrictions

What Is Control Stock?

- Stock owned by affiliates (officers, directors, >10% owners)
- Not restricted by acquisition method
- Restricted by WHO owns it
- Sale limitations apply regardless of how acquired

SIE Exam Tip: Restricted = how you got it; Control = who you are.

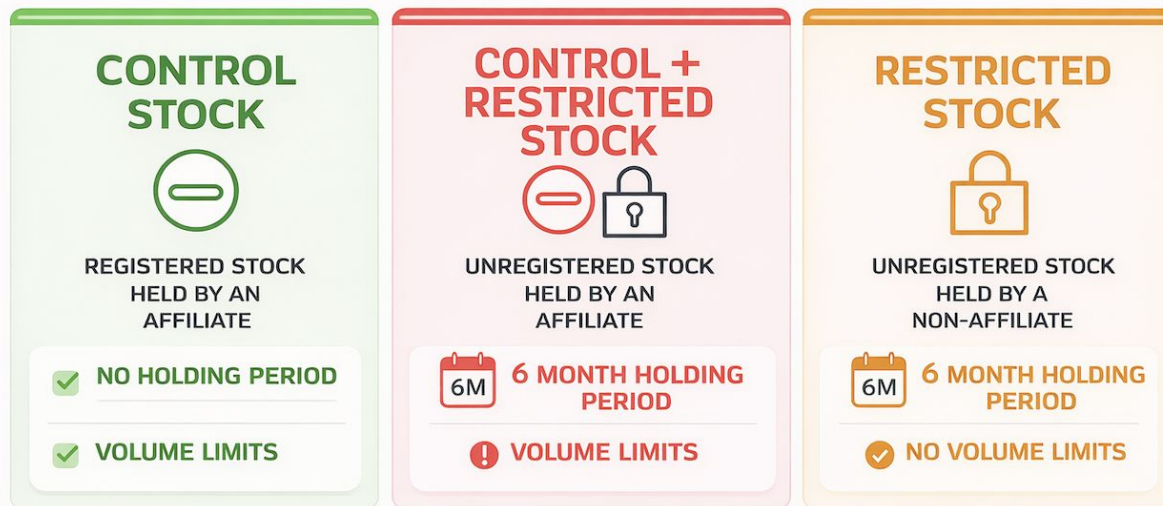
Rule 144 Overview

- SEC rule allowing resale of restricted/control stock
- Restricted stock: 6-month (reporting) or 12-month (non-reporting) holding period
- Volume limits: Can't dump all at once
- Filing requirements: Form 144 for larger sales

SIE Exam Tip: Rule 144 = "safe harbor" for selling restricted/control stock.

Key Takeaways

- Restricted stock = limited by how acquired (private placement)
- Control stock = limited by who owns it (affiliates)
- Rule 144 provides path to sell both



Introduction to Debt Securities

Learning Objectives:

- Define debt securities and the creditor/borrower relationship
- Identify major categories of debt issuers
- Contrast debt with equity at a high level

What Is a Debt Security?

- Represents a LOAN from investor to issuer
- Issuer is the borrower; investor is the creditor
- Issuer promises to repay principal + interest
- Fixed income = predictable payment stream

SIE Exam Tip: Debt = lending; Equity = owning. Foundational distinction.

The Creditor Relationship

- Bondholders are creditors, NOT owners
- No voting rights in company decisions
- No share in profits beyond stated interest
- Priority over stockholders in bankruptcy

Example: You buy a corporate bond - you've lent money to the company, not bought ownership.

Who Issues Debt Securities?

- U.S. Treasury: T-Bills, T-Notes, T-Bonds
- Federal Agencies/GSEs: Fannie Mae, Freddie Mac
- Corporations: Corporate bonds
- Municipalities: State and local governments
- Each has different risk and tax treatment

Why Issuers Use Debt

- Raise capital without diluting ownership
- Interest payments are tax-deductible (corporations)
- Fixed obligation—must repay regardless of profits
- Cheaper than equity when rates are low

SIE Exam Tip: Interest is tax-deductible for corporations; dividends are not.

Why Investors Buy Debt

- Predictable income stream
- Lower risk than equity (generally)
- Priority in bankruptcy
- Diversification from stocks

Debt vs. Equity Preview

Feature	Debt (Bonds)	Equity (Stock)
Relationship	Creditor	Owner
Returns	Interest (fixed)	Dividends (variable)
Bankruptcy priority	Higher	Lower
Upside potential	Capped	Unlimited

SIE Exam Tip: This comparison is tested repeatedly—memorize it.

Key Takeaways

- Debt securities = loans to issuers
- Investor is creditor with priority over equity
- Major issuers: Treasury, agencies, corporations, municipalities
- Fixed income = more predictable but capped returns

Bond Fundamentals

Learning Objectives:

- Define par value, coupon rate, and maturity
- Explain why bonds trade at premium or discount
- Understand the inverse relationship between price and yield

Bond Terminology Basics

- Par value (face value): Amount repaid at maturity (typically \$1,000)
- Coupon rate: Stated annual interest rate (fixed at issuance)
- Maturity date: When principal is repaid
- Coupon payment: Dollar amount of interest paid

Example: \$1,000 par, 5% coupon = \$50 annual interest

Par, Premium & Discount

- At par: Bond price = \$1,000 (100% of face)
- Premium: Bond price > \$1,000 (above par)
- Discount: Bond price < \$1,000 (below par)
- Price expressed as percentage of par (e.g., 98 = \$980)

SIE Exam Tip: Bond prices quoted as percentage of par, not dollars.

Why Bonds Trade at Premium

- Coupon rate > current market rates
- Investors pay MORE to get higher interest
- Example: 6% coupon when market rates are 4%
- Bond is "worth more" than par

Example: You'd pay extra for a bond paying 6% when new bonds only pay 4%.

Why Bonds Trade at Discount

- Coupon rate < current market rates
- Investors pay LESS because interest is below market
- Example: 3% coupon when market rates are 5%
- Discount compensates for lower income

SIE Exam Tip: Discount bonds have coupon BELOW market rates.

The Seesaw Relationship

- Interest rates $\uparrow \rightarrow$ Bond prices $\downarrow$
- Interest rates $\downarrow \rightarrow$ Bond prices $\uparrow$
- Inverse relationship—ALWAYS
- Existing bonds must compete with new issue rates

SIE Exam Tip: The inverse price/yield relationship is one of the most tested concepts.

Visualizing the Seesaw



- Think of a seesaw: one side goes up, other goes down
- This applies to ALL fixed-rate bonds

Coupon Rate Is Fixed

- Coupon rate set at issuance—never changes
- A 5% bond always pays 5% of par (\$50/year)
- What changes is the PRICE and YIELD
- Don't confuse coupon rate with yield

SIE Exam Tip: Coupon rate = fixed. Yield = changes with price.

Bond Quote Example

- Bond quoted at 102.50
- Price = $102.50\% \times \$1,000 = \$1,025$
- Trading at premium (above par)
- Implies coupon rate > current market rates

Bond Fundamentals: Key Takeaways

- Par = \$1,000; Coupon = stated interest rate
- Premium = price above par; Discount = below par
- Rates up → Prices down (inverse relationship)
- Coupon rate is fixed; yield fluctuates with price

Bond Yields: Current Yield, YTM & YTC

Learning Objectives:

- Calculate and interpret current yield
- Understand Yield to Maturity (YTM) and Yield to Call (YTC)
- Know the yield hierarchy for premium and discount bonds

Why Yields Matter

- Yield = true return on bond investment
- Coupon rate alone doesn't tell the whole story
- Must account for price paid

Nominal Yield (Coupon Rate)

- Stated interest rate on the bond
- Annual interest $\div$ Par value
- Example: \$50 interest $\div$ \$1,000 par = 5%
- Does NOT reflect actual return if price $\neq$ par

SIE Exam Tip: Nominal yield = coupon rate. Simplest yield measure.

Current Yield

- Annual interest $\div$ Current market price
- Reflects income return at today's price
- Formula: Current Yield = Annual Coupon $\div$ Market Price

Example: \$50 coupon $\div$ \$950 price = 5.26% current yield

SIE Exam Tip: Current yield only measures income—ignores capital gain/loss at maturity.

Current Yield Examples

Price	Annual Coupon	Current Yield
\$1,000 (par)	\$50	5.00%
\$950 (discount)	\$50	5.26%
\$1,050 (premium)	\$50	4.76%

SIE Exam Tip: Discount bond → current yield > coupon rate. Premium → current yield < coupon rate.

Yield to Maturity (YTM)

- Total return if held to maturity
- Includes coupon payments AND capital gain/loss
- Most comprehensive yield measure
- Assumes reinvestment at same rate

SIE Exam Tip: YTM = best measure of total return for bonds held to maturity.

YTM and Price Relationship

- Discount bond: $YTM > \text{Current Yield} > \text{Coupon Rate}$
- Premium bond: $YTM < \text{Current Yield} < \text{Coupon Rate}$
- Par bond: $YTM = \text{Current Yield} = \text{Coupon Rate}$

Memory trick: Discount = yields go UP (ascending). Premium = yields go DOWN (descending).

SIE Exam Tip: The yield hierarchy is heavily tested—memorize the order.

Yield to Call (YTC)

- Return if bond is called before maturity
- Relevant only for callable bonds
- Usually calculated to first call date
- Lower than YTM for premium bonds (shorter time)

Example: 10-year bond callable in 5 years—YTC assumes redemption at year 5.

SIE Exam Tip: Callable bonds trading at premium: $YTC < YTM$ (worst case for investor).

Which Yield to Use?

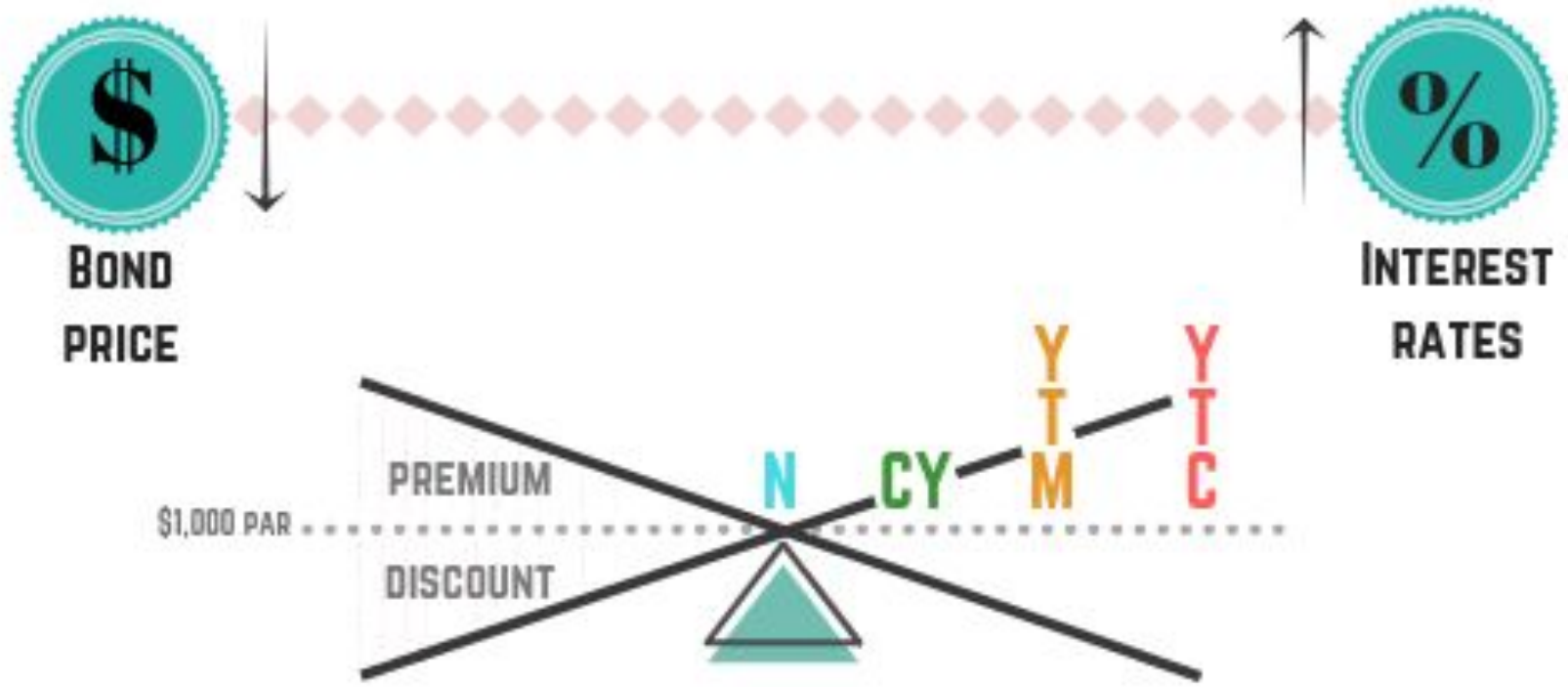
Scenario	Use This Yield
Compare income only	Current Yield
Hold to maturity	YTM
Callable bond at premium	YTC (worst case)
Quick comparison	Current Yield

SIE Exam Tip: For callable premium bonds, YTC is the most conservative (lowest) yield.

Yield Summary Table

Yield Type	Formula/Concept	Includes
Nominal (Coupon)	$\text{Coupon} \div \text{Par}$	Interest only
Current Yield	$\text{Coupon} \div \text{Price}$	Interest only
YTM	Total return to maturity	Interest + gain/loss
YTC	Total return to call	Interest + gain/loss

The Bond Seesaw: Price vs. Yield



Bond Yield - Key Takeaways

- Nominal yield = coupon rate (fixed)
- Current yield = coupon $\div$ market price
- YTM = total return held to maturity
- YTC = return if called early
- Discount: $YTM > CY > \text{Coupon}$; Premium: opposite

U.S. Treasury Securities

Learning Objectives:

- Distinguish T-Bills, T-Notes, and T-Bonds by maturity
- Explain Treasury auction mechanics
- Understand why Treasuries are considered risk-free

Overview of Treasury Securities

- Issued by U.S. Department of Treasury
- Backed by "full faith and credit" of U.S. government
- Considered virtually default-free
- Benchmark for other interest rates
- SIE Exam Tip: Treasuries = safest debt securities (no credit risk)

Treasury Bills (T-Bills)

- Maturity: 4, 8, 13, 17, 26, or 52 weeks (≤ 1 year)
- Sold at discount to face value
- No coupon payments (zero-coupon)
- Return = difference between purchase price and par

Example: Buy T-Bill at \$9,800, receive \$10,000 at maturity = \$200 return

SIE Exam Tip: T-Bills = short-term, discount securities, no interest payments

Treasury Notes (T-Notes)

- Maturity: 2, 3, 5, 7, or 10 years
- Pay semi-annual interest (coupon)
- Sold at, above, or below par
- Most actively traded Treasury securities

SIE Exam Tip: T-Notes = medium-term with semi-annual interest

Treasury Bonds (T-Bonds)

- Maturity: 20 or 30 years
- Pay semi-annual interest
- Longest-term Treasury securities
- Most sensitive to interest rate changes

SIE Exam Tip: Longer maturity = more interest rate risk. T-Bonds have the most

Treasury Maturity Summary

- T-Bills: ≤ 1 year maturity, Discount (no interest)
- T-Notes: 2–10 years maturity, Semi-annual interest
- T-Bonds: 20–30 years maturity, Semi-annual interest

Memory trick: Bills are short, Notes in the middle, Bonds are long

Treasury Auctions

- Competitive bid: Specify yield; may not be filled
- Non-competitive bid: Accept auction yield; always filled
- Non-competitive limited to \$10 million per auction
- Individual investors typically use non-competitive

SIE Exam Tip: Non-competitive = guaranteed to get securities.
Competitive = may be shut out

Treasury Taxation

- Interest taxable at FEDERAL level
- Exempt from state and local taxes
- Advantage for investors in high-tax states
- Capital gains still taxable at all levels

SIE Exam Tip: Treasury interest = federal tax only. State/local exempt

Why Treasuries Are "Risk-Free"

- No credit/default risk (government can print money)
- Highly liquid (active secondary market)
- BUT still have interest rate risk
- "Risk-free" refers only to default risk

SIE Exam Tip: Treasuries have interest rate risk—just no credit risk

U.S. Treasuries - Key Takeaways

- T-Bills: ≤ 1 year, discount, no coupon
- T-Notes: 2–10 years, semi-annual coupon
- T-Bonds: 20–30 years, semi-annual coupon
- Auctions: competitive vs. non-competitive bids
- Tax: federal only, state/local exempt

STRIPS & TIPS

Learning Objectives:

- Explain how STRIPS are created and their characteristics
- Describe how TIPS protect against inflation
- Understand phantom income taxation on zero-coupon bonds

What Are STRIPS?

- Separate Trading of Registered Interest and Principal Securities
- Created by "stripping" T-Notes/Bonds into components
- Each coupon payment and principal becomes separate security
- Zero-coupon Treasury securities

SIE Exam Tip: STRIPS = zero-coupon Treasuries created from regular bonds.

How STRIPS Work

- 10-year T-Note with semi-annual payments creates:
- 20 interest "strips" (each coupon payment)
- 1 principal "strip" (face value at maturity)
- Total: 21 separate securities, each sold at discount

Example: Buy a principal strip for \$500 today, receive \$1,000 in 10 years.

STRIPS Characteristics

- Sold at deep discount to par
- No periodic interest payments
- Return = difference between purchase price and par
- High interest rate sensitivity (long duration)

SIE Exam Tip: Zero-coupon bonds have the MOST interest rate risk for their maturity.

STRIPS Taxation (Phantom Income)

- Must pay taxes on "accrued" interest annually
- Even though no cash received until maturity
- Called "phantom income"
- Better suited for tax-deferred accounts (IRAs)

SIE Exam Tip: Phantom income = taxed on interest you haven't received.
Common exam topic.

Who Buys STRIPS?

- Investors with specific future cash needs
- Pension funds matching future liabilities
- College savings (known future expense)
- Tax-deferred account holders

Example: Need exactly \$50,000 in 15 years? Buy STRIPS maturing then.

What Are TIPS?

- Treasury Inflation-Protected Securities
- Principal adjusts with Consumer Price Index (CPI)
- Protects purchasing power from inflation
- Issued in 5, 10, and 30-year maturities

SIE Exam Tip: TIPS protect against inflation; STRIPS do not.

How TIPS Work

- Fixed coupon rate applied to adjusted principal
- Inflation: Principal increases → higher interest payment
- Deflation: Principal decreases (but floor at original par)
- At maturity: receive greater of adjusted or original principal

Example: \$1,000 TIPS, 2% coupon. After 3% inflation: $\$1,030 \times 2\% = \20.60 interest.

TIPS Taxation

- Interest payments taxed annually (like regular bonds)
- Principal adjustments ALSO taxed annually
- Another form of phantom income
- Better in tax-deferred accounts

SIE Exam Tip: TIPS have phantom income from inflation adjustments—taxed before received.

Key Takeaways

- STRIPS: Zero-coupon Treasuries, deep discount, phantom income
- TIPS: Principal adjusts with inflation, protects purchasing power
- Both have phantom income taxation
- STRIPS = most interest rate sensitive
- TIPS = best inflation hedge among Treasuries

Agency & Mortgage-Backed Securities

Learning Objectives:

- Distinguish government agencies from GSEs
- Explain mortgage-backed securities and prepayment risk
- Understand CMO basics

What Are Agency Securities?

- Issued by federal agencies or government-sponsored enterprises (GSEs)
- Not directly backed by U.S. government (except GNMA)
- Lower yield than corporate, higher than Treasury
- Includes debt securities and mortgage-backed securities

Government Agencies vs. GSEs

- Government Agency: GNMA (Ginnie Mae) — Full faith & credit backing
- GSE: FNMA (Fannie Mae), FHLMC (Freddie Mac) — Implicit backing (not guaranteed)

SIE Exam Tip: Only GNMA has explicit government guarantee. Fannie/Freddie do not.

What Are Mortgage-Backed Securities (MBS)?

- Pools of mortgages packaged into securities
- Investors receive "pass-through" of mortgage payments
- Monthly payments include principal AND interest
- Created by GNMA, FNMA, FHLMC

Example: 1,000 home mortgages pooled together; investors buy shares of the pool.

Prepayment Risk

- Homeowners can pay off mortgages early (refinance)
- When rates DROP, prepayments INCREASE
- Investors receive principal back sooner than expected
- Must reinvest at lower rates

SIE Exam Tip: Prepayment risk is the key MBS risk—increases when rates fall.

Extension Risk

- Opposite of prepayment risk
- When rates RISE, prepayments SLOW
- Principal returned more slowly than expected
- Investor stuck with below-market yields longer

SIE Exam Tip: Rates up = extension risk. Rates down = prepayment risk.

Collateralized Mortgage Obligations (CMOs)

- MBS restructured into "tranches" (slices)
- Different tranches have different prepayment exposure
- Shorter tranches absorb prepayments first
- Allows investors to choose risk level

SIE Exam Tip: CMOs don't eliminate prepayment risk—they redistribute it among tranches.

CMO Tranches

- Sequential tranches: Paid in order (A, B, C, etc.)
- PAC tranches: More predictable cash flows
- Z-tranches: Receive no payments until others paid
- Complexity = suitability concerns

SIE Exam Tip: CMOs are complex; not suitable for unsophisticated investors.

Other Asset-Backed Securities (ABS)

- Backed by non-mortgage loans
- Auto loans, Credit card receivables, Student loans
- Similar structure to MBS
- Prepayment risk varies by collateral type

MBS - Key Takeaways

- Agency securities: lower risk than corporate, higher than Treasury
- Only GNMA has full government guarantee
- MBS: monthly pass-through of mortgage payments
- Prepayment risk when rates fall; extension risk when rates rise
- CMOs redistribute (don't eliminate) prepayment risk

Corporate Bonds

Learning Objectives:

- Distinguish secured from unsecured corporate bonds
- Explain callable and convertible features
- Understand credit ratings and default risk

Overview of Corporate Bonds

- Debt issued by corporations
- Higher yield than government bonds
- Higher credit risk (default possible)
- Interest taxable at all levels (federal, state, local)

SIE Exam Tip: Corporate bond interest is fully taxable—no exemptions.

Secured Bonds

- Backed by specific collateral
- If issuer defaults, bondholders claim collateral
- Mortgage bonds: Real estate collateral
- Equipment trust certificates: Equipment (planes, trains)
- Collateral trust bonds: Securities as collateral

SIE Exam Tip: Secured = specific assets backing the bond. Lower risk than unsecured.

Unsecured Bonds (Debentures)

- Backed only by issuer's creditworthiness
- No specific collateral
- Debenture: General unsecured bond
- Subordinated debenture: Paid AFTER other debt in bankruptcy

SIE Exam Tip: Subordinated = junior = paid last among bondholders.

Corporate Bond Priority

Priority (Highest to Lowest):

1. Secured bonds
2. Unsecured bonds (debentures)
3. Subordinated debentures
4. Preferred stock
5. Common stock

SIE Exam Tip: Bankruptcy priority order is heavily tested.

Callable Bonds

- Issuer can redeem before maturity
- Called when interest rates DROP
- Issuer refinances at lower rate
- Call premium: Extra amount paid to call

Example: 6% bond callable at 102. Issuer calls when rates drop to 4%.

SIE Exam Tip: Callable benefits ISSUER—call risk hurts investor.

Convertible Bonds

- Can be converted to common stock
- Conversion at bondholder's option
- Offers upside if stock rises
- Lower coupon than non-convertible

Example: \$1,000 bond convertible to 40 shares. If stock rises to \$30, conversion value = \$1,200.

SIE Exam Tip: Convertible gives equity upside; investor accepts lower yield.

Credit Ratings

- Issued by Moody's, S&P, Fitch
- Measure probability of default
- Investment grade: BBB/Baa and above
- High-yield (junk): Below BBB/Baa
- AAA/Aaa = Highest quality | BBB/Baa = Lowest investment grade | BB/Ba = Highest junk

SIE Exam Tip: BBB/Baa is the dividing line between investment grade and junk.

Credit Risk vs. Yield

- Lower rating = higher yield (compensation for risk)
- Higher rating = lower yield
- "Risk/return tradeoff"
- Ratings can change (upgrades/downgrades)

SIE Exam Tip: Junk bonds offer higher yields because of higher default risk.

Corporate Bonds - Key Takeaways

- Secured bonds: backed by collateral (safest)
- Debentures: unsecured; subordinated = junior
- Callable: issuer can redeem (benefits issuer)
- Convertible: can become stock (benefits investor)
- BBB/Baa = dividing line for investment grade

Municipal Bonds

Learning Objectives:

- Distinguish General Obligation from Revenue bonds
- Explain the tax advantages of municipal bonds
- Calculate tax-equivalent yield

What Are Municipal Bonds?

- Issued by state and local governments
- Fund public projects (roads, schools, hospitals)
- Key benefit: tax-exempt interest
- Lower yields but tax advantages offset

Tax Advantages

- Interest exempt from FEDERAL income tax
- Often exempt from state/local tax if bought in-state
- Triple tax-free: Federal + state + local exempt
- Capital gains still taxable

SIE Exam Tip: Muni interest = federal tax-exempt. Most attractive to high-tax-bracket investors.

General Obligation (GO) Bonds

- Backed by "full faith and credit" of issuer
- Repaid from general tax revenues
- Voter approval typically required
- Considered safer than revenue bonds

SIE Exam Tip: GO bonds = backed by taxing power. Very safe if issuer has strong tax base.

Types of GO Backing

- Unlimited tax GO: Can raise taxes as needed
- Limited tax GO: Tax increase capped
- Double-barreled: Backed by taxes AND revenue source
- Moral obligation: State may (not must) support

SIE Exam Tip: Unlimited tax GO is safest; moral obligation provides no legal guarantee.

Revenue Bonds

- Backed by revenue from specific project
- Examples: toll roads, airports, hospitals
- No taxing power behind them
- Riskier than GO bonds

Example: Airport revenue bond repaid from landing fees and terminal rents.

SIE Exam Tip: Revenue bonds depend on project success—if project fails, bonds may default.

Revenue Bond Protections

- Rate covenant: Issuer promises to maintain rates
- Flow of funds: Order in which revenues applied
- Debt service reserve: Funds set aside for payments
- Protective covenants reduce risk

Tax-Equivalent Yield

- Compares muni yield to taxable bond yield
- Formula: Tax-Equivalent Yield = Muni Yield $\div$ (1 – Tax Rate)

Example: 3% muni, 32% tax bracket: $3\% \div (1 - 0.32) = 3\% \div 0.68 = 4.41\%$
equivalent

SIE Exam Tip: Higher tax bracket = greater muni benefit. Know the formula concept.

Who Should Buy Munis?

- High-income/high-tax-bracket investors
- Investors in high-tax states (for in-state bonds)
- Those seeking tax-free income
- NOT suitable for tax-deferred accounts (no benefit)

SIE Exam Tip: Munis in an IRA = bad idea. Tax exemption wasted.

Municipal Bond Risks

- Credit risk: Issuer may default (less common)
- Interest rate risk: Prices fall when rates rise
- Call risk: Many munis are callable
- Liquidity risk: Less active market than Treasuries

Muni Bonds - Key Takeaways

- Munis: tax-exempt interest (federal; often state)
- GO bonds: backed by taxes (safer)
- Revenue bonds: backed by project revenue (riskier)
- Best for high-tax-bracket investors
- Don't put munis in tax-deferred accounts

Money Market Instruments

Learning Objectives:

- Identify short-term debt instruments
- Distinguish between the major money market securities
- Understand their role in portfolio liquidity

What Is the Money Market?

- Market for short-term debt (≤ 1 year maturity)
- High liquidity, low risk
- Used for cash management
- Lower yields than longer-term bonds

SIE Exam Tip: Money market = short-term, liquid, low risk

Treasury Bills (Review)

- U.S. government short-term debt
- Maturities: 4 weeks to 52 weeks
- Sold at discount, mature at par
- Safest money market instrument

Commercial Paper

- Short-term unsecured corporate debt
- Maturity: 1 to 270 days (typically)
- Issued by large, creditworthy corporations
- Sold at discount

SIE Exam Tip: Commercial paper = unsecured corporate IOU. Only creditworthy issuers

Banker's Acceptances (BAs)

- Short-term debt guaranteed by bank
- Used in international trade
- Buyer's bank "accepts" obligation to pay
- Trades at discount in secondary market

Example: Importer's bank guarantees payment to exporter—creates BA

SIE Exam Tip: Banker's acceptances = international trade financing

Negotiable CDs

- Bank-issued time deposits
- Large denominations (\$100,000+)
- Fixed interest rate and maturity
- Can be traded in secondary market

SIE Exam Tip: Negotiable CDs can be sold before maturity (unlike regular CDs)

Repurchase Agreements (Repos)

- Short-term borrowing using securities as collateral
- Dealer sells securities, agrees to repurchase
- Overnight or term (days to weeks)
- Used by dealers and institutions

Example: Bank sells Treasuries overnight, buys back next day at slightly higher price

SIE Exam Tip: Repo = sell now, buy back later. Reverse repo = buy now, sell back later

Federal Funds

- Overnight loans between banks
- Used to meet reserve requirements
- Fed funds rate: Key short-term interest rate
- Banks with excess reserves lend to those short

SIE Exam Tip: Fed funds rate is the benchmark short-term rate set by bank lending

Money Market Summary

- T-Bills — Issuer: U.S. Treasury — Key Feature: Safest
- Commercial Paper — Issuer: Corporations — Key Feature: Unsecured, short
- Banker's Acceptances — Issuer: Banks — Key Feature: Trade finance
- Negotiable CDs — Issuer: Banks — Key Feature: Tradeable deposits
- Repos — Issuer: Dealers — Key Feature: Collateralized borrowing
- Fed Funds — Issuer: Banks — Key Feature: Overnight interbank

Money Markets - Key Takeaways

- Money market = short-term, liquid, low-risk
- T-Bills safest; commercial paper = corporate
- Repos = short-term collateralized borrowing
- Fed funds rate = benchmark for short-term rates
- All mature in ≤ 1 year

Bond Risks

Learning Objectives:

- Identify and explain all major bond risks
- Understand how different bonds face different risks
- Connect risks to investor suitability

Overview of Bond Risks

- All bonds carry risk (yes, even Treasuries)
- Different bonds = different risk profiles
- Understanding risks is key to suitability
- SIE tests risk identification heavily

Interest Rate Risk

- Risk that rates rise, causing bond prices to fall
- Affects ALL fixed-rate bonds
- Longer maturity = more interest rate risk
- Lower coupon = more interest rate risk

SIE Exam Tip: Zero-coupon, long-term bonds have the MOST interest rate risk

Credit Risk (Default Risk)

- Risk issuer fails to pay interest or principal
- Measured by credit ratings
- Treasuries: No credit risk
- Corporate/Muni: Varies by issuer

SIE Exam Tip: Higher yield = higher credit risk. Risk/return tradeoff.

Call Risk

- Risk that issuer redeems bond early
- Happens when rates DROP
- Investor loses higher-yielding bond
- Must reinvest at lower rates

SIE Exam Tip: Call risk and reinvestment risk go together—both hurt when rates fall.

Reinvestment Risk

- Risk of reinvesting cash flows at lower rates
- Affects coupon payments and called bonds
- Higher coupon bonds have more reinvestment risk
- Zero-coupon bonds have NO reinvestment risk (no coupons)

SIE Exam Tip: Zero-coupon = no reinvestment risk (but maximum interest rate risk)

Inflation Risk (Purchasing Power)

- Risk that inflation erodes real returns
- Fixed payments buy less over time
- TIPS protect against this
- Long-term bonds most vulnerable

Example: 3% bond with 4% inflation = -1% real return

Liquidity Risk

- Risk of not selling quickly at fair price
- Treasuries = highly liquid
- Munis, small corporate issues = less liquid
- Wider bid/ask spread indicates low liquidity

Prepayment Risk (MBS)

- Risk of early principal return
- Rates DROP → prepayments INCREASE
- Must reinvest at lower rates
- Unique to mortgage-backed securities

Extension Risk (MBS)

- Opposite of prepayment risk
- Rates RISE → prepayments SLOW
- Stuck with below-market yields longer
- Principal tied up longer than expected

Risk Summary by Bond Type

- Treasury: Interest rate only
- Corporate: Credit, interest rate, call
- Municipal: Credit, interest rate, call, liquidity
- MBS/CMO: Prepayment, extension, interest rate
- Zero-Coupon: Interest rate (max), phantom income

Bond Risks - Key Takeaways

- Interest rate risk: all fixed-rate bonds (longer = more)
- Credit risk: not Treasuries; yes corporate/muni
- Call + reinvestment: both hurt when rates fall
- Prepayment (MBS): rates fall → prepay faster
- Zero-coupon: max interest rate risk, zero reinvestment risk

Introduction to Derivatives

Learning Objectives:

- Define derivative securities and explain the concept of underlying assets
- Distinguish hedging from speculation
- Connect warrants (from equity) to the broader derivatives concept

What Is a Derivative?

- Financial contract whose value is "derived" from something else
- The "something else" = underlying asset
- Underlying can be: stocks, bonds, commodities, currencies, indexes
- Derivatives don't represent ownership—they represent rights or obligations

SIE Exam Tip: Derivative = value derived from underlying asset.
Foundation concept.

Types of Underlying Assets

- Equity: Individual stocks, stock indexes (S&P 500)
- Debt: Interest rates, Treasury bonds
- Commodities: Oil, gold, wheat, corn
- Currencies: EUR/USD, foreign exchange rates
- Other: Volatility indexes (VIX)

Example: A call option on Apple stock—Apple is the underlying asset.

Main Types of Derivatives

- Options: Right (not obligation) to buy or sell
- Futures: Obligation to buy or sell at future date
- Forwards: Like futures, but customized and OTC
- Warrants: Long-term options issued by companies

SIE Exam Tip: Options = rights. Futures/Forwards = obligations. Critical distinction.

Hedging vs. Speculation

- Hedging: Reducing existing risk
 - Example: Farmer locks in wheat price to protect against price drop
- Speculation: Taking on risk for potential profit
 - Example: Trader bets oil prices will rise

SIE Exam Tip: Know the purpose difference—hedging protects, speculation profits from movement.

Hedging Example

- You own 100 shares of XYZ stock at \$50
- Worried about price decline
- Buy a put option (right to sell at \$50)
- If stock drops to \$40, your put protects you
- This is hedging—reducing downside risk

Speculation Example

- You don't own any XYZ stock
- You believe price will rise from \$50 to \$70
- Buy a call option (right to buy at \$50)
- If right, you profit; if wrong, you lose premium
- This is speculation—betting on direction

Warrants Revisited (Bridge from Equity)

- Long-term options to buy stock at fixed price
- Issued by the company itself (not OCC)
- Usually attached to bonds or preferred as "sweetener"
- Exercise price typically above market when issued
- Duration: years (vs. months for listed options)

SIE Exam Tip: Warrants are company-issued derivatives. Listed options are standardized through OCC.

Why Derivatives Matter

- Risk management tools for investors and businesses
- Can amplify gains (leverage)
- Can also amplify losses
- Require understanding before using
- SIE tests basic concepts, not complex strategies

Derivatives - Key Takeaways

- Derivatives derive value from underlying assets
- Options = right; Futures/Forwards = obligation
- Hedging = reduce risk; Speculation = seek profit
- Warrants are long-term, company-issued options

SIE focuses on definitions and basic distinctions

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Options Basics: Calls & Puts

Learning Objectives:

- Define call and put options and their rights
- Explain premium, strike price, and expiration
- Understand intrinsic value vs. time value

What Is an Option?

- Contract giving the RIGHT (not obligation) to buy or sell
- Right exists for limited time (until expiration)
- Buyer pays premium for this right
- Two types: Calls and Puts

SIE Exam Tip: Option buyers have RIGHTS. Option sellers have OBLIGATIONS.

Call Options

- Right to BUY the underlying at the strike price
- Buyer expects price to RISE
- Pays premium for this right
- Profitable when market price > strike price

Example: Buy a call with \$50 strike. Stock rises to \$60. You can buy at \$50 (below market).

SIE Exam Tip: Call = right to BUY. "Call it away" from someone.

Put Options

- Right to SELL the underlying at the strike price
- Buyer expects price to FALL
- Pays premium for this right
- Profitable when market price < strike price

Example: Buy a put with \$50 strike. Stock falls to \$40. You can sell at \$50 (above market).

SIE Exam Tip: Put = right to SELL. "Put it to" someone.

Key Options Terminology

- Premium: Price paid for the option
- Strike (Exercise) Price: Price at which you can buy/sell
- Expiration Date: Last day option can be exercised
- Underlying: The asset the option controls
- Contract Size: Standard = 100 shares per contract

SIE Exam Tip: 1 option contract = 100 shares. Premium quoted per share.

Premium Example

- Option premium quoted at \$3
- Contract controls 100 shares
- Total cost = $\$3 \times 100 = \300
- This is the maximum the buyer can lose
- This is the maximum the seller can gain (from premium)

In-the-Money, At-the-Money, Out-of-the-Money

For CALL options (Strike \$50):

- In-the-money (ITM): Market > Strike (Stock at \$55)
- At-the-money (ATM): Market = Strike (Stock at \$50)
- Out-of-the-money (OTM): Market < Strike (Stock at \$45)

SIE Exam Tip: ITM calls have intrinsic value. OTM calls have zero intrinsic value.

In/At/Out-of-the-Money for Puts

For PUT options (Strike \$50):

- In-the-money (ITM): $\text{Market} < \text{Strike}$ (Stock at \$45)
- At-the-money (ATM): $\text{Market} = \text{Strike}$ (Stock at \$50)
- Out-of-the-money (OTM): $\text{Market} > \text{Strike}$ (Stock at \$55)

SIE Exam Tip: Puts are opposite of calls—ITM when market is BELOW strike.

Intrinsic Value vs. Time Value

Intrinsic value: Real, immediate value if exercised now

- Call: Market price – Strike price (if positive)
- Put: Strike price – Market price (if positive)
- Time value: Extra premium for time remaining
- Formula: Premium = Intrinsic Value + Time Value

SIE Exam Tip: OTM options have ZERO intrinsic value—only time value.

Intrinsic Value Example

Call option: Strike \$50, Premium \$7, Stock at \$54

Calculation:

- Intrinsic value = $\$54 - \$50 = \$4$
- Time value = $\$7 - \$4 = \$3$
- Premium (\$7) = Intrinsic (\$4) + Time (\$3)

SIE Exam Tip: Know how to calculate intrinsic value for both calls and puts.

American vs. European Options

- American style: Can exercise ANY time before expiration
- European style: Can exercise ONLY at expiration
- Most listed stock options are American style
- Index options are often European style

SIE Exam Tip: American = anytime; European = expiration only. American is more flexible.

Options Clearing Corporation (OCC)

- Standardizes listed options contracts
- Guarantees performance (counterparty risk eliminated)
- Sets strike prices and expiration dates
- Facilitates exercise and assignment

SIE Exam Tip: OCC is the issuer and guarantor of all listed options.

Options Basics - Key Takeaways

- Call = right to buy; Put = right to sell
- Premium = price paid; Strike = exercise price
- ITM = has intrinsic value; OTM = no intrinsic value
- Premium = Intrinsic value + Time value
- American options can be exercised anytime; European only at expiration

Options Positions: Buyers vs. Writers

Learning Objectives:

- Distinguish long (buyer) from short (writer) positions
- Calculate maximum gain, maximum loss, and breakeven
- Understand the risk/reward profile of each position

Four Basic Options Positions

- Long Call: Buy a call (right to buy)
- Short Call: Sell/write a call (obligation to sell)
- Long Put: Buy a put (right to sell)
- Short Put: Sell/write a put (obligation to buy)

SIE Exam Tip: "Long" = buyer = pays premium = has rights. "Short" = seller = receives premium = has obligations.

Long Call Position

- Outlook: Bullish (expects price to rise)
- Pays: Premium
- Right: To buy at strike price
- Max Loss: Premium paid (limited)
- Max Gain: Unlimited (stock can rise indefinitely)
- Breakeven: Strike price + Premium

Example: Buy call, strike \$50, premium \$3. Breakeven = \$53. Max loss = \$300 (\$3 × 100).



Short Call Position (Writing a Call)

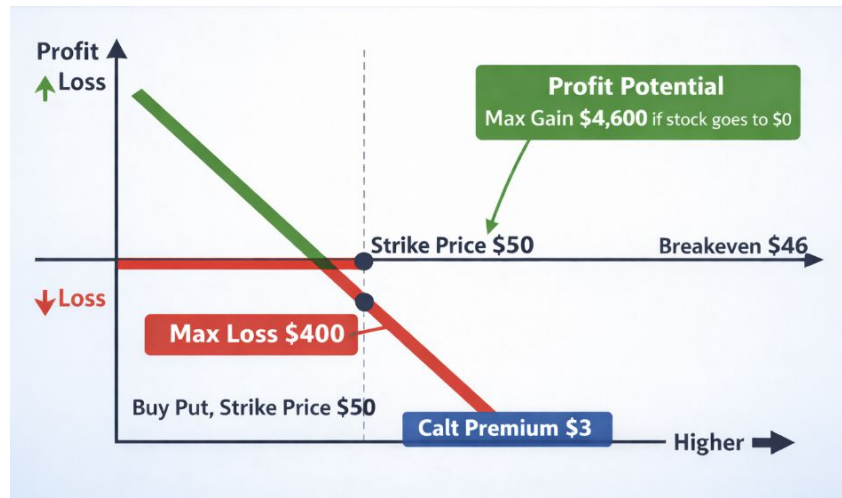
- Outlook: Neutral to bearish (expects price stable or down)
- Receives: Premium
- Obligation: To sell at strike price if assigned
- Max Gain: Premium received (limited)
- Max Loss: UNLIMITED (stock can rise indefinitely)

SIE Exam Tip: Short call = UNLIMITED risk. Very dangerous uncovered ("naked").

Long Put Position

- Outlook: Bearish (expects price to fall)
- Pays: Premium
- Right: To sell at strike price
- Max Loss: Premium paid (limited)
- Max Gain: Strike price - Premium (stock can fall to zero)
- Breakeven: Strike price - Premium

Example: Buy put, strike \$50, premium \$4. Breakeven = \$46. Max gain if stock goes to \$0 = \$4,600.



Short Put Position (Writing a Put)

- Outlook: Neutral to bullish (expects price stable or up)
- Receives: Premium
- Obligation: To buy at strike price if assigned
- Max Gain: Premium received (limited)
- Max Loss: Strike price - Premium (if stock goes to zero)

SIE Exam Tip: Short put = must buy stock if assigned. Risk = owning stock at inflated price.

Summary of Four Positions

Position	Outlook	Max Gain	Max Loss	Breakeven
Long Call	Bullish	Unlimited	Premium	Strike + Premium
Short Call	Bearish/Neutral	Premium	Unlimited	Strike + Premium
Long Put	Bearish	Strike – Premium	Premium	Strike – Premium
Short Put	Bullish/Neutral	Premium	Strike – Premium	Strike – Premium

SIE Exam Tip: Memorize this table - it's heavily tested.

Buyers vs. Writers Summary

Aspect	Buyer (Long)	Writer (Short)
Pays/Receives	Pays premium	Receives premium
Has	Rights	Obligations
Wants option to	Be exercised (ITM)	Expire worthless (OTM)
Time decay	Enemy (hurts)	Friend (helps)

SIE Exam Tip: Time decay benefits writers - options lose value as expiration approaches.

Assignment and Exercise

- Exercise: Buyer uses their right
- Assignment: Writer must fulfill obligation
- Only ITM options are typically exercised
- OTM options expire worthless

Example: You wrote a call, strike \$50. Stock at \$60. Buyer exercises; you must sell at \$50.

Covered vs. Uncovered (Naked)

- Covered: Writer owns underlying stock
- Example: Own stock + write call = covered call (limited additional risk)
- Uncovered (Naked): Writer doesn't own underlying
- Example: Write call without owning stock = unlimited risk

SIE Exam Tip: Naked options = highest risk. Not suitable for most retail investors.

Options Positions Key Takeaways

- Short put: gain premium, lose strike minus premium
- Breakeven for calls: $\text{Strike} + \text{Premium}$
- Breakeven for puts: $\text{Strike} - \text{Premium}$
- Time decay hurts buyers, helps writers
- Only ITM options are typically exercised

Futures & Forwards

Learning Objectives:

- Define futures and forward contracts
- Distinguish exchange-traded futures from OTC forwards
- Understand the obligation nature vs. options' rights

What Are Futures and Forwards?

- Contracts to buy or sell an asset at a future date
- Price agreed today, delivery later
- Key difference from options: These are OBLIGATIONS, not rights
- Both parties MUST perform

SIE Exam Tip: Futures/Forwards = obligations. Options = rights. Critical distinction.

Futures Contracts

- Standardized contracts traded on exchanges
- Fixed quantities, quality, delivery dates
- Marked to market daily (gains/losses settled daily)
- Clearinghouse guarantees performance

Example: Crude oil futures: 1,000 barrels, specific grade, delivery in March

Futures Contract Features

- Standardized: Set contract sizes and terms
- Exchange-traded: CME, CBOT, NYMEX
- Margin required: Deposit as performance guarantee
- Marked to market: Daily settlement of gains/losses

SIE Exam Tip: Futures = standardized + exchange + clearinghouse + daily settlement

Forward Contracts

- Customized contracts traded over-the-counter (OTC)
- Terms negotiated between parties
- NOT marked to market (settled at end)
- No clearinghouse guarantee—higher counterparty risk

Example: Company contracts to exchange €1M for USD in 90 days at agreed rate

Forwards Contract Features

- Customized: Tailored to parties' needs
- OTC-traded: Private negotiation
- No margin: Credit risk between parties
- Settlement at end: No daily marking

SIE Exam Tip: Forwards = customized + OTC + counterparty risk + settlement at end

Futures vs. Forwards Comparison

Aspect	Futures	Forwards
Trading venue	Exchange	OTC
Standardization	Yes	No (customized)
Counterparty risk	None (clearinghouse)	Yes
Marked to market	Daily	No (end settlement)
Liquidity	High	Low
Regulation	High	Low

SIE Exam Tip: This comparison is tested on the exam

Futures vs. Options

Aspect	Futures	Options
Buyer has	Obligation	Right
Seller has	Obligation	Obligation
Premium paid	No (margin deposit)	Yes
Can expire worthless	No	Yes
Risk profile	Symmetric	Asymmetric

SIE Exam Tip: Both parties obligated in futures. Only seller obligated in options.

Hedging with Futures

- Producers: Lock in selling price
Example: Farmer sells wheat futures to guarantee price
- Consumers: Lock in buying price
Example: Airline buys oil futures to fix fuel costs
- Reduces uncertainty for both parties

Speculating with Futures

- Traders bet on price direction
- Long futures: Profit if price rises
- Short futures: Profit if price falls
- Leverage amplifies gains AND losses—can lose more than initial margin

SIE Exam Tip: Futures speculation is high-risk due to leverage

Margin in Futures

- Initial margin: Deposit required to open position
- Maintenance margin: Minimum balance required
- Margin call: Must add funds if below maintenance
- NOT a loan (unlike stock margin)—it's a performance bond

SIE Exam Tip: Futures margin = good faith deposit, not borrowing

Common Futures Contracts

- Commodities: Oil, gold, wheat, corn, cattle
- Financials: Treasury bonds, Eurodollars
- Currencies: EUR/USD, JPY/USD
- Indexes: S&P 500 futures, Nasdaq futures

SIE Exam Tip: Futures exist on many underlying assets—not just commodities

Futures and Forward - Key Takeaways

- Futures = standardized, exchange-traded, daily settlement
- Forwards = customized, OTC, settlement at end
- Both are OBLIGATIONS (unlike options which are rights)
- Futures have clearinghouse (no counterparty risk)
- Forwards have counterparty risk—both used for hedging and speculation

Mutual Funds: Structure, Pricing & Share Classes

Learning Objectives:

- Explain how mutual funds are structured and regulated
- Calculate and interpret Net Asset Value (NAV)
- Distinguish share classes and understand sales charges and fees

What Is a Mutual Fund?

- Pooled investment from many investors
- Professionally managed portfolio
- Open-end investment company under Investment Company Act of 1940
- Investors buy/sell shares directly from the fund
- Shares redeemed at NAV (not traded on exchange)

SIE Exam Tip: Mutual fund = open-end = redeemable shares at NAV. Key classification.

How Mutual Funds Work

- Fund issues unlimited shares (open-end)
- Investors buy shares → fund grows
- Investors redeem shares → fund shrinks
- Manager invests according to fund objective

SIE Exam Tip: Open-end funds continuously issue and redeem shares.

Net Asset Value (NAV)

- Price per share of a mutual fund
- Calculated once daily after market close (4:00 PM ET)
- Formula: $NAV = (\text{Total Assets} - \text{Total Liabilities}) \div \text{Shares Outstanding}$

Example: Fund has \$100M assets, \$2M liabilities, 5M shares. $NAV = (\$100M - \$2M) \div 5M = \$19.60$

SIE Exam Tip: NAV calculated once daily at market close. Know the formula.

Public Offering Price (POP)

- Price investors pay to buy shares
- $POP = NAV + \text{Sales Charge}$
- Sales charge (load) compensates the seller
- No-load funds: $POP = NAV$

Example: $NAV = \$20$, 5% front-end load. $POP = \$20 \div (1 - 0.05) = \21.05

SIE Exam Tip: POP is what you pay. NAV is what the shares are worth.
Difference = sales charge.

Types of Sales Charges (Loads)

- Front-end (Class A): Charged at purchase; reduces initial investment
- Back-end/CDSC (Class B): Charged at redemption; decreases over time
- Level load (Class C): Ongoing higher 12b-1 fees; continuous cost
- No-load: Never charged; no sales charge

SIE Exam Tip: Front-end = pay upfront. Back-end (CDSC) = pay when you sell.
Level = ongoing.

Share Class Comparison

- Class A: Front-end load (3-5%), low 12b-1 fees (0.25%), breakpoints available
- Class B: No front-end, back-end load (declines), high 12b-1 (1%), converts to A
- Class C: No front-end, small back-end (1 year), high 12b-1 (1%), best for short-term

SIE Exam Tip: Class A = breakpoints available. Class B = converts to A. Class C = level load.

12b-1 Fees

- Distribution and marketing fees charged by fund
- Paid from fund assets (reduces returns)
- Maximum: 1% of average net assets annually
- 0.25% = can still be called "no-load"
- Covers advertising, compensation to brokers, shareholder services

SIE Exam Tip: 12b-1 fees = ongoing marketing costs. Max 1%. Under 0.25% = can be "no-load."

Other Mutual Fund Fees

- Management fee: Paid to investment adviser
- Expense ratio: Total annual operating expenses as % of assets
- Redemption fee: Short-term trading penalty (not a sales load)
- Exchange fee: Fee to switch between funds in same family

SIE Exam Tip: Expense ratio = total annual cost. Lower = generally better for investor.

Breakpoints

- Volume discounts on front-end sales charges
- Larger investment = lower percentage load
- Letter of Intent (LOI): Commit to future purchases for immediate discount (13 months)
- Rights of Accumulation (ROA): Current holdings count toward breakpoint

Example: Breakpoint at \$50,000 reduces load from 5% to 4%. Invest \$50,000 = save \$500.

SIE Exam Tip: Selling just below a breakpoint is a violation ("breakpoint selling").

Breakpoint Example

- Under \$25,000: 5.00% sales charge
- \$25,000 – \$49,999: 4.50% sales charge
- \$50,000 – \$99,999: 4.00% sales charge
- \$100,000 – \$249,999: 3.00% sales charge
- \$250,000+: 2.00% sales charge

SIE Exam Tip: Breakpoint selling (keeping customer just below threshold) = prohibited practice.

Types of Mutual Funds by Objective

- Growth funds: Capital appreciation (higher risk)
- Income funds: Current income from dividends/interest (moderate risk)
- Balanced funds: Growth + income (moderate risk)
- Index funds: Match benchmark passively (varies by index)
- Money market funds: Stability and liquidity (low risk)

SIE Exam Tip: Know fund types for suitability questions.

Diversified vs. Non-Diversified Funds

- Diversified Fund (75-5-10 rule):
 - 75% of assets must be diversified; no more than 5% in any one issuer
 - No more than 10% of any issuer's voting securities
- Non-Diversified: Can concentrate in fewer securities; higher risk/reward

SIE Exam Tip: 75-5-10 rule defines "diversified" under Investment Company Act of 1940.

Mutual Fund Taxation

- Dividends and capital gains distributed to shareholders
- Shareholders taxed on distributions (even if reinvested)
- Qualified dividends: Lower tax rate; Short-term gains: Ordinary income rate
- Fund itself not taxed if it distributes 90%+ of income

SIE Exam Tip: Shareholders pay tax on distributions even if reinvested—common test point.

Mutual Funds - Key Takeaways

- Mutual funds = open-end investment companies, redeemable at NAV
- NAV calculated daily = $(\text{Assets} - \text{Liabilities}) \div \text{Shares}$
- Share classes: A (front-end), B (back-end), C (level load)
- 12b-1 fees = marketing costs, max 1%
- Breakpoints reduce sales charges on larger investments
- 75-5-10 rule = diversified fund requirements

ETFs: Exchange-Traded Funds

Learning Objectives:

- Explain ETF structure and how they trade
- Compare ETFs to mutual funds
- Understand index-based and actively managed ETFs

What Is an ETF?

- Exchange-Traded Fund
- Trades on exchange like a stock
- Holds basket of securities (stocks, bonds, commodities)
- Combines features of mutual funds and stocks
- Most track an index (passive management)

SIE Exam Tip: ETF = trades on exchange (like stock) but holds diversified portfolio (like mutual fund).

How ETFs Trade

- Trade throughout the day at market price
- Buy/sell through broker (not from fund directly)
- Market orders, limit orders apply
- Price can differ from NAV (premium or discount)
- Bid/ask spread applies (trading cost)

SIE Exam Tip: ETFs trade at market price, which may differ from NAV. Mutual funds always trade at NAV.

ETF Creation and Redemption

- Authorized Participants (APs): Large institutions that create/redeem shares
- Creation: AP delivers basket of securities - receives ETF shares
- Redemption: AP delivers ETF shares - receives basket of securities
- In-kind transfers: Securities, not cash (tax efficient)
- Keeps ETF price close to NAV through arbitrage

SIE Exam Tip: Creation/redemption mechanism keeps ETF price near NAV.
Tax efficient.

ETF vs. Mutual Fund Comparison

Feature	ETF	Mutual Fund
Trading	Throughout day	Once daily at NAV
Pricing	Market price (real-time)	NAV (end of day)
Minimum investment	1 share	Often \$1,000+
Sales loads	None (but commissions may apply)	Often yes (A, B, C)
Tax efficiency	Generally higher	Generally lower

SIE Exam Tip: This comparison is heavily tested. Know all differences.

ETF Advantages

- Intraday trading: Buy/sell anytime market is open
- Lower expense ratios: Typically cheaper than mutual funds
- Tax efficiency: In-kind redemptions minimize capital gains
- Transparency: Holdings disclosed daily
- No minimum investment: Buy as little as 1 share

SIE Exam Tip: ETF advantages: intraday trading, tax efficiency, lower costs.

ETF Disadvantages

- Bid/ask spread: Cost to trade
- Commissions: May apply (though many brokers offer free ETF trades)
- Premium/discount to NAV: May pay more or less than underlying value
- Tracking error: May not perfectly match index
- Less suitable for: Dollar-cost averaging small amounts

SIE Exam Tip: ETF costs include spread and potential premium/discount to NAV.

Types of ETFs

Type	Description	Example
Index ETFs	Track market index	S&P 500 ETF
Sector ETFs	Focus on specific industry	Technology ETF
Bond ETFs	Hold fixed-income securities	Corporate bond ETF
Commodity ETFs	Track commodity prices	Gold ETF
Inverse ETFs	Profit when index falls	Short S&P 500 ETF
Leveraged ETFs	Magnified returns (2x, 3x)	2x S&P 500 ETF

SIE Exam Tip: Leveraged and inverse ETFs = high risk, not suitable for long-term investors.

Leveraged and Inverse ETFs

- Leveraged: Seek 2x or 3x daily return of index
- Inverse: Seek opposite of daily return
- Reset daily: Compounding can cause unexpected results
- Short-term trading tools: Not suitable for buy-and-hold

Example: 2x S&P 500 ETF: If S&P rises 1%, ETF aims for 2%

SIE Exam Tip: Leveraged/inverse ETFs reset daily - long-term results may differ significantly.

Index Funds vs. Index ETFs

- Trading: Index Mutual Fund = End of day | Index ETF = Intraday
- Minimum: Index Mutual Fund = Often \$1,000+ | Index ETF = 1 share
- Tax efficiency: Index Mutual Fund = Lower | Index ETF = Higher
- Expense ratio: Both low, but ETFs usually lower
- Automatic investing: Mutual Fund = Easy | ETF = Harder (whole shares)

SIE Exam Tip: Both track indexes, but ETFs trade intraday and are more tax efficient.

ETF Suitability

- Suitable For: Cost-conscious investors, Tax-sensitive accounts
- Suitable For: Active traders, Diversification seekers
- Not Suitable For: Frequent small investments, Automatic investment plans
- Not Suitable For: Investors unfamiliar with trading
- Leveraged/Inverse ETFs: Short-term only, NOT for long-term investors

SIE Exam Tip: Match ETF type to investor needs. Leveraged ETFs = short-term only.

ETFs - Key Takeaways

- ETFs trade on exchanges throughout the day at market price
- NAV and market price may differ (premium/discount)
- Creation/redemption by APs keeps price near NAV
- Generally lower costs and more tax efficient than mutual funds
- Most track indexes (passive management)
- Leveraged/inverse ETFs = high risk, daily reset, short-term only

Closed-End Funds & UITs

Learning Objectives:

- Distinguish closed-end funds from open-end (mutual) funds
- Explain premium and discount to NAV
- Understand Unit Investment Trust structure

Investment Company Types

The Investment Company Act of 1940 defines three types:

- Open-end (Mutual funds): Unlimited shares, redeemable at NAV
- Closed-end funds: Fixed shares, trade on exchange
- UITs: Fixed portfolio, self-liquidating

SIE Exam Tip: Know all three types and their key distinguishing features.

What Is a Closed-End Fund?

- Fixed number of shares issued through IPO
- Shares trade on exchange (like stocks)
- Fund does NOT redeem shares
- Buy/sell from other investors (secondary market)
- Professionally managed portfolio

SIE Exam Tip: Closed-end = fixed shares, trades on exchange, NOT redeemable from fund.

Closed-End Fund Pricing

- Trade at market price (determined by supply/demand)
- Market price often differs from NAV
- Premium: Market price $>$ NAV (investors pay more than assets worth)
- Discount: Market price $<$ NAV (investors pay less than assets worth)

Example: NAV = \$20, Market price = \$18. Fund trades at 10% discount.

SIE Exam Tip: Closed-end funds often trade at discount to NAV.

Why Premiums and Discounts Exist

Premium (Price > NAV):

- Strong demand, popular manager, unique strategy, limited supply

Discount (Price < NAV):

- Weak demand, poor performance, market pessimism, illiquid holdings

SIE Exam Tip: Premium/discount reflects investor sentiment, not just underlying asset value.

Closed-End vs. Open-End Comparison

- Shares: Closed-End = Fixed (IPO) | Mutual Fund = Unlimited
- Trading: Closed-End = Exchange (intraday) | Mutual Fund = Directly with fund
- Pricing: Closed-End = Market price | Mutual Fund = NAV only
- Premium/Discount: Closed-End = Yes | Mutual Fund = No
- Leverage: Closed-End = Yes | Mutual Fund = Limited

SIE Exam Tip: Closed-end can trade at premium/discount and use leverage. Mutual funds cannot.

Closed-End Fund Features

- Can use leverage: Borrow money or issue preferred shares
- May invest in illiquid securities: No redemption pressure
- Income focus: Many pay regular distributions
- IPO pricing: Initial shares sold with sales charge
- No continuous offering: Cannot buy directly from fund after IPO

SIE Exam Tip: Leverage increases both return potential AND risk in closed-end funds.

What Is a Unit Investment Trust (UIT)?

- Fixed portfolio of securities
- Not actively managed (no portfolio manager)
- Has a termination date (self-liquidating)
- Trust issues units to investors
- Holds securities until maturity or termination

SIE Exam Tip: UIT = fixed, unmanaged portfolio with termination date.

UIT Structure

- Sponsor: Creates the trust and selects securities
- Trustee: Holds securities and administers the trust
- Units: Sold to investors (redeemable from trust)
- No management fee (but sales charges apply)
- Income: Passed through to unit holders

SIE Exam Tip: UIT has no investment adviser/manager - portfolio is fixed at creation.

Types of UITs

- Equity UIT: Holds stocks, often 1-5 years duration
- Bond UIT: Holds bonds until they mature
- Municipal bond UIT: Holds muni bonds, tax-exempt income

Example: Bond UIT holds 20 corporate bonds maturing in 10 years. Trust terminates when bonds mature.

SIE Exam Tip: Bond UITs terminate when underlying bonds mature.

UIT vs. Mutual Fund vs. Closed-End

- Management: UIT = None (fixed) | Mutual Fund = Active | Closed-End = Active
- Portfolio changes: UIT = No | Mutual Fund = Yes | Closed-End = Yes
- Termination date: UIT = Yes | Mutual Fund = No | Closed-End = No
- Trades at NAV: UIT = Yes | Mutual Fund = Yes | Closed-End = No

SIE Exam Tip: UIT = fixed portfolio, termination date, no manager.

UIT Costs

- Sales charge: Front-end load (typically 1-5%)
- Creation and development fee: One-time cost
- No ongoing management fee (portfolio is fixed)
- Annual operating expenses: Administrative costs
- Generally lower ongoing costs than mutual funds

SIE Exam Tip: UITs have sales charges but no management fees.

Closed-End Funds & UITs - Key Takeaways

- Closed-end funds: Fixed shares, trade on exchange, premium/discount to NAV
- Closed-end can use leverage; mutual funds cannot
- UITs: Fixed portfolio, no manager, termination date
- UITs are redeemable (unlike closed-end)
- All three are investment companies under 1940 Act

Other Pooled Investments

Learning Objectives:

- Understand hedge fund basics at SIE level
- Explain fund of funds structure
- Know key Investment Company Act provisions

Beyond Traditional Investment Companies

- Some pooled investments fall outside 1940 Act
- Hedge funds, private equity: For sophisticated investors
- Different rules and investor protections
- SIE tests basic awareness, not details

SIE Exam Tip: Know that alternatives exist and have different rules—details are Series 7/66.

Hedge Funds Overview

- Private investment pools for wealthy investors
- Not registered under Investment Company Act
- Use complex strategies: leverage, short selling, derivatives
- Accredited investors or Qualified Purchasers only
- Less regulated, less transparent

SIE Exam Tip: Hedge funds = unregistered, accredited investors only, complex strategies.

Hedge Fund Characteristics

Feature	Hedge Fund
Registration	Not registered (1940 Act exempt)
Investors	Accredited/Qualified Purchasers
Liquidity	Limited (lock-up periods)
Fees	High ("2 and 20": 2% management + 20% performance)
Strategies	Long/short, leverage, derivatives
Transparency	Limited disclosure

SIE Exam Tip: "2 and 20" fee structure = 2% of assets + 20% of profits.

Accredited Investor Definition

- Income test: \$200,000 individual / \$300,000 joint for past 2 years
- Net worth test: \$1 million excluding primary residence
- Institutional investors: Banks, insurance companies, pension funds
- Certain professionals: Licensed securities professionals

SIE Exam Tip: Know income (\$200K/\$300K) and net worth (\$1M) thresholds.

Qualified Purchaser

- Higher standard than accredited investor
- Individual: \$5 million+ in investments
- Institution: \$25 million+ in investments
- Required for certain private funds
- Allows funds to avoid most 1940 Act requirements

SIE Exam Tip: Qualified Purchaser = \$5M investments (higher than Accredited Investor).

Fund of Funds

- Mutual fund or hedge fund that invests in other funds
- Diversification across multiple managers/strategies
- Layer of fees: Both underlying funds AND fund of funds charge fees
- May provide access to otherwise inaccessible funds

SIE Exam Tip: Fund of funds = extra layer of fees. Consider total cost.

Fund of Funds Structure

Investor → Fund of Funds → Multiple Underlying Funds

- Investor pays fund of funds fee
- Fund of funds pays underlying fund fees
- Double layer of expenses reduces returns

SIE Exam Tip: Fund of funds has two layers of fees—may not be cost efficient.

Investment Company Act of 1940 (Key Provisions)

Provision	Requirement
Registration	Investment companies must register with SEC
Prospectus	Must provide to investors before sale
Board composition	At least 40% independent directors
Custody	Assets held by qualified custodian
Diversification	75-5-10 rule for diversified funds
Pricing	Daily NAV calculation required

SIE Exam Tip: 40% independent directors is a key governance requirement.

Investment Company Prohibited Activities

- Cannot issue senior securities (debt) except as permitted
- Cannot purchase on margin
- Cannot engage in short selling (generally)
- Cannot have affiliated transactions without approval
- Must maintain minimum liquidity

SIE Exam Tip: Open-end funds face restrictions closed-end funds don't.

Comparing Pooled Investments

Type	Registration	Investors	Liquidity	Fees
Mutual Fund	Registered	Anyone	Daily	Moderate
ETF	Registered	Anyone	Intraday	Low
Closed-End	Registered	Anyone	Intraday	Moderate
UIT	Registered	Anyone	Redeemable	Low ongoing
Hedge Fund	Exempt	Accredited	Limited	High

SIE Exam Tip: Registered = retail investors. Exempt = sophisticated investors only.

Pooled Investments - Key Takeaways

- Hedge funds: unregistered, accredited investors, high fees, complex strategies
- Accredited investor: \$200K/\$300K income OR \$1M net worth
- Qualified Purchaser: \$5M in investments (higher standard)
- Fund of funds: invests in other funds, double layer of fees
- Investment Company Act of 1940: registration, prospectus, 40% independent directors

Introduction to Investment Risks

Importance of understanding risks

Types of Investment Risks (Overview)

- Capital Risk - Risk of losing principal
Example: Startup investment failure
- Credit Risk - Applicable to bonds
Example: Default by a company with poor credit
- Currency Risk - Impact on international investments
Example: Euro weakening vs. dollar
- Inflationary Risk - Erosion of purchasing power
Example: Negative real returns due to inflation

Types of Investment Risks (Overview)

- **Rising rates impact bonds**
Example: Reinvesting at lower rates
- **Liquidity Risk - Difficulty selling quickly**
Example: Real estate vs. stocks
- **Market/Systematic Risk - Impact of market-wide declines**
Example: 2008 financial crisis
- **Non-Systematic Risk - Company or industry-specific risk**
Example: CEO scandal

Types of Investment Risks (Overview)

- **Political Risk - Changes in government or policy**
Example: Regulation impacting energy companies
- **Prepayment Risk - Common in mortgage-backed securities**
Example: Early loan repayment

Mitigating Risk Strategies

- Diversification: Spread investments across different asset classes.
- Rebalancing: Adjust portfolio periodically to maintain desired asset allocation.
- Hedging: Use financial instruments to protect against potential losses.

Direct Participation Programs (DPPs)

Learning Objectives:

- Define DPPs and explain pass-through taxation
- Distinguish general partners from limited partners
- Identify key DPP risks and suitability concerns

What Is a DPP?

- Pooled investment in business ventures (real estate, oil/gas, equipment leasing)
- Pass-through taxation: Income/losses flow directly to investors' tax returns
- No corporate-level taxation
- Typically structured as limited partnerships

SIE Exam Tip: DPPs "pass through" income AND losses—key tax benefit.

Limited Partnership Structure

- General Partner (GP): Unlimited liability, runs the business
- Limited Partner (LP): Limited liability, passive investor
- GPs make decisions and bear full liability
- LPs contribute capital, cannot participate in management
- If LP participates in management → loses limited liability protection

SIE Exam Tip: Limited partners = limited liability + limited control. Heavily tested.

Types of DPPs

- Real Estate: Commercial property, development
- Oil & Gas: Exploration, drilling, income programs
- Equipment Leasing: Aircraft, computers, machinery
- Agriculture: Farming operations

SIE Exam Tip: Know that DPPs exist in multiple industries—not just real estate.

Pass-Through Taxation

- DPP itself pays no federal income tax
- Profits AND losses pass to partners' individual returns
- Investors receive Schedule K-1 (not Form 1099)
- Losses can offset other passive income

SIE Exam Tip: K-1 = partnership income. 1099 = other investment income.
Know the difference.

DPP Risks

- Illiquidity: No secondary market; hard to sell
- Business risk: Venture may fail
- Legislative risk: Tax laws can change
- Lack of control: LPs cannot make decisions

SIE Exam Tip: Illiquidity is the #1 DPP risk tested on the exam.

Suitability Considerations

- Suitable for: High-net-worth investors seeking tax benefits
- NOT suitable for: Those needing liquidity or capital preservation
- Accredited investor requirements often apply
- Long holding periods (5–10+ years typical)

SIE Exam Tip: DPPs are NOT suitable for investors who may need quick access to funds.

Tenants in Common (TIC)

- Direct ownership of undivided fractional interest in property
- Each owner holds title independently
- Can sell or transfer share without other owners' consent
- Income and expenses shared proportionally
- Often used for 1031 exchanges

SIE Exam Tip: TIC = direct ownership. LP = ownership through partnership.

Tax Benefits vs. Economic Merit

- DPPs must have economic viability beyond tax benefits
- IRS scrutinizes programs sold primarily for tax write-offs
- "Abusive tax shelters" = penalties for investor AND promoter
- Tax benefits are secondary; investment merit comes first

SIE Exam Tip: Economic merit must be primary purpose—not just tax avoidance.

DPPs - Key Takeaways

- DPPs = pass-through taxation (no double taxation)
- GP = unlimited liability + control; LP = limited liability + passive
- Primary risks: illiquidity, business risk, legislative risk
- Suitable for high-net-worth, long-term investors
- Must have economic merit beyond tax benefits

Real Estate Investment Trusts (REITs)

Learning Objectives:

- Define REITs and explain their tax treatment
- Distinguish equity, mortgage, and hybrid REITs
- Compare REITs to DPPs

What Is a REIT?

- Company that owns/operates income-producing real estate
- Allows investment in real estate without direct ownership
- Structured as corporations or trusts
- Income from rent, mortgage interest, or property sales

SIE Exam Tip: REITs provide real estate exposure without buying property directly.

REIT Tax Treatment

- Must distribute at least 90% of taxable income to shareholders
- If 90%+ distributed → REIT pays no corporate income tax
- Dividends taxed to shareholders as ordinary income (not qualified)
- Avoids double taxation like DPPs

SIE Exam Tip: 90% distribution requirement = key testable fact.

Types of REITs

- Equity REIT: Owns properties (apartments, malls, offices) → Income from rent
- Mortgage REIT: Invests in mortgages and MBS → Income from interest
- Hybrid REIT: Both properties and mortgages → Rent + Interest

SIE Exam Tip: Equity REITs = property ownership. Mortgage REITs = debt investments.

REIT Trading Categories

- Listed (Public): Trades on exchange → High liquidity, SEC reporting
- Non-Listed (Public): SEC registered, no exchange → Low liquidity
- Private: Accredited investors only → Lowest liquidity

SIE Exam Tip: Listed REITs = liquid. Non-listed = illiquid despite SEC registration.

REIT Risks

- Interest rate risk: Rising rates hurt property values and borrowing costs
- Market risk: REIT prices fluctuate with the market (listed REITs)
- Liquidity risk: Non-listed and private REITs are hard to sell
- Economic risk: Vacancy rates and tenant defaults affect income

SIE Exam Tip: Interest rate risk is significant—REITs often use leverage.

REITs vs. DPPs Comparison

- Liquidity: REITs = High (if listed) | DPPs = Very low
- Tax form: REITs = 1099 | DPPs = K-1
- Pass-through losses: REITs = No | DPPs = Yes
- 90% distribution: REITs = Required | DPPs = Not required

SIE Exam Tip: REITs = 1099, no loss pass-through. DPPs = K-1, losses flow through.

Suitability Considerations

- Listed REITs: Suitable for income-seeking investors wanting liquidity
- Non-listed REITs: Higher yield potential, but illiquid
- Real estate exposure without property management hassle
- NOT suitable for those prioritizing capital appreciation over income

SIE Exam Tip: Match REIT type to investor's liquidity needs.

REITs - Key Takeaways

- REITs = real estate investment without direct ownership
- Must distribute 90%+ of income → no corporate tax
- Equity REITs own property; Mortgage REITs hold debt
- Listed REITs = liquid; Non-listed = illiquid
- Dividends taxed as ordinary income (not qualified)

Retirement Accounts: The Big Picture

Learning Objectives:

- Understand the purpose of tax-advantaged retirement accounts
- Distinguish tax-deferred from tax-free growth
- Differentiate qualified from non-qualified plans

Why Retirement Accounts Matter

- Encourage long-term saving for retirement
- Tax advantages accelerate growth
- Different account types for different situations
- SIE tests rules, limits, and tax treatment

SIE Exam Tip: Focus on tax treatment and contribution/distribution rules—not specific investment advice.

Tax-Deferred vs. Tax-Free Growth

- Tax-Deferred: Pre-tax contributions, tax-free growth, taxed as ordinary income on withdrawal
- Tax-Free: After-tax contributions, tax-free growth, tax-free withdrawals (if qualified)
- Tax-deferred examples: Traditional IRA, 401(k)
- Tax-free examples: Roth IRA, Roth 401(k)

SIE Exam Tip: Tax-deferred = taxed on withdrawal. Tax-free (Roth) = taxed on contribution.

Qualified vs. Non-Qualified Plans

- Qualified: Meets IRS requirements, tax advantages (401(k), IRA, 403(b), pension)
- Non-Qualified: No special tax treatment, flexible (deferred compensation, executive plans)
- Qualified plans have contribution limits and rules
- Non-qualified plans have fewer restrictions but no tax benefits

SIE Exam Tip: Qualified = tax advantages + IRS rules. Non-qualified = flexible but no tax break.

Contribution vs. Distribution Taxation

- Traditional IRA/401(k): Tax-deductible contribution, taxed as ordinary income on distribution
- Roth IRA/401(k): Not deductible contribution, tax-free distribution (if qualified)
- Non-qualified annuity: After-tax contribution, earnings taxed as ordinary income

SIE Exam Tip: Know when taxes are paid—contribution OR distribution, not both.

Early Withdrawal Penalty

- Withdrawals before age 59½ generally trigger 10% penalty
- Penalty is IN ADDITION to ordinary income tax
- Applies to qualified plans and IRAs
- Exceptions exist (disability, first home, education, etc.)

SIE Exam Tip: 59½ is the magic age. Before = 10% penalty (with exceptions).

Required Minimum Distributions (RMDs)

- Must begin withdrawals at age 73 (for most)
- Applies to Traditional IRAs, 401(k)s, other qualified plans
- Roth IRAs: No RMDs during owner's lifetime
- Failure to take RMD = 25% penalty on amount not withdrawn

SIE Exam Tip: Traditional = RMDs required. Roth IRA = NO RMDs. Key distinction.

Retirement Accounts - Key Takeaways

- Tax-deferred = pay taxes on withdrawal (Traditional)
- Tax-free = pay taxes on contribution, never again (Roth)
- Qualified plans have tax advantages but strict rules
- Early withdrawal (before 59½) = 10% penalty
- RMDs begin at 73 (except Roth IRAs)

Individual Retirement Accounts (IRAs)

Learning Objectives:

- Distinguish Traditional from Roth IRAs
- Know contribution limits and eligibility rules
- Understand RMDs, early withdrawals, and rollovers

What Is an IRA?

- Individual Retirement Account
- Personal retirement savings with tax advantages
- Two main types: Traditional and Roth
- Owned by individual (not employer-sponsored)
- Can hold stocks, bonds, mutual funds, ETFs

SIE Exam Tip: IRA = individual account. 401(k) = employer plan. Different rules.

Traditional IRA Overview

- Contributions: May be tax-deductible (depends on income/employer plan)
- Growth: Tax-deferred
- Withdrawals: Taxed as ordinary income
- RMDs: Required starting at age 73
- Best for: Those expecting lower tax bracket in retirement

SIE Exam Tip: Traditional IRA = tax break NOW, pay taxes LATER.

Roth IRA Overview

- Contributions: NOT tax-deductible (after-tax money)
- Growth: Tax-free
- Withdrawals: Tax-free (if qualified)
- RMDs: None during owner's lifetime
- Income limits: High earners may not be eligible

SIE Exam Tip: Roth = pay taxes NOW, withdrawals are TAX-FREE.

Traditional vs. Roth IRA Comparison

Growth	Tax-deferred	Tax-free
Withdrawals	Taxed	Tax-free
RMDs	Yes (age 73)	No
Income limits	None for contributions	Yes
Early withdrawal	10% penalty + tax	Contributions: no penalty

IRA Contribution Limits (2024/2025)

- Annual limit: \$7,000 (under age 50)
- Catch-up: Additional \$1,000 if age 50+ (total \$8,000)
- Limit applies to TOTAL IRA contributions (Traditional + Roth combined)
- Must have earned income to contribute
- Cannot contribute more than earned income

SIE Exam Tip: Know the limits and that they're combined across IRA types.

IRA Eligibility

- Traditional IRA: Anyone with earned income (no income limit)
- Roth IRA: Income limits apply; high earners may be ineligible
- Spousal IRA: Non-working spouse can contribute based on working spouse's income

SIE Exam Tip: Roth has income limits; Traditional does not (deductibility may be limited).

Early Withdrawal Rules

- Withdrawals before 59½ = 10% penalty + ordinary income tax
- Exceptions to penalty (still owe tax): Disability
- First-time home purchase (\$10,000 lifetime)
- Qualified education expenses, unreimbursed medical expenses
- Death (beneficiary)

SIE Exam Tip: Know the common exceptions—they're tested.

Roth IRA Withdrawal Rules

- Contributions: Can withdraw anytime, tax-free, penalty-free
- Earnings: Tax-free only if "qualified distribution"
- Qualified distribution requires: Account open 5+ years AND
- Age 59½, disability, death, or first home (\$10,000)
- Withdrawals come out: contributions first, then earnings

SIE Exam Tip: Roth contributions = always accessible. Earnings = 5-year rule + 59½.

Rollovers and Transfers

- Direct rollover/transfer: Funds move institution-to-institution = No tax, no penalty
- Indirect rollover: You receive check, must deposit within 60 days
- 20% withholding on indirect; must replace to avoid tax
- 60-day rule: Must complete indirect rollover in 60 days or = taxable distribution

SIE Exam Tip: Direct = no tax. Indirect = 60-day deadline and 20% withheld.

SEP and SIMPLE IRAs

- SEP IRA: Self-employed, small business; employer contributes only; high limits
- SIMPLE IRA: Small business (≤ 100 employees); employee + employer contributions
- Both are employer-sponsored but use IRA structure
- Higher contribution limits than regular IRAs

SIE Exam Tip: SEP = employer only. SIMPLE = both contribute.

Key Takeaways

- Traditional: deductible contributions, taxed withdrawals, RMDs at 73
- Roth: after-tax contributions, tax-free withdrawals, no RMDs
- Contribution limit: \$7,000 (\$8,000 if 50+)—combined total
- Early withdrawal = 10% penalty (exceptions exist)
- Direct rollover = no tax; Indirect = 60-day rule

Employer-Sponsored Retirement Plans

Learning Objectives:

- Explain 401(k), 403(b), and 457 plan basics
- Understand employer matching, vesting, and loans
- Know rollover rules when leaving an employer

Overview of Employer Plans

- Retirement savings through your employer
- Defined contribution: You contribute; balance depends on investments
- Defined benefit: Employer promises specific retirement income (pension)
- Most common today: 401(k) and 403(b)

SIE Exam Tip: Defined contribution = your risk. Defined benefit = employer's promise.

401(k) Plans

- Who offers: Private (for-profit) employers
- Contributions: Employee pre-tax (or Roth option)
- Employer match: Common (e.g., 50% of first 6%)
- Limits (2024): \$23,000 employee; \$69,000 total with employer
- Catch-up (50+): Additional \$7,500

Example: Earn \$100,000, contribute 6% (\$6,000). Employer matches 50% = \$3,000. Total = \$9,000.

SIE Exam Tip: 401(k) = private employers. 403(b) = nonprofits/schools.

403(b) Plans

- Who offers: Nonprofits, schools, hospitals, government
- Similar to 401(k) in structure and limits
- Investment options: Often limited to annuities and mutual funds
- Also called: Tax-Sheltered Annuity (TSA)

SIE Exam Tip: 403(b) = nonprofits/education. Same limits as 401(k).

457 Plans

- Who offers: State/local government and some nonprofits
- Key advantage: No 10% early withdrawal penalty (government 457)
- Can contribute to BOTH 457 and 401(k)/403(b)
- Same contribution limits as 401(k)

SIE Exam Tip: Government 457 plans = NO early withdrawal penalty. Unique feature.

Employer Plan Comparison

Plan	Employer Type	Early Withdrawal Penalty
401(k)	Private/for-profit	Yes (10%)
403(b)	Nonprofit/education	Yes (10%)
457 (govt)	Government	No
Pension	Any	N/A (defined benefit)

SIE Exam Tip: Only government 457 plans avoid the 10% penalty.

Employer Matching

- “Free money” from employer
- Common formula: 50% or 100% match up to X% of salary
- Match may be subject to vesting schedule
- Employee contributions = always 100% vested

Example: Employer matches 100% of first 3%. You contribute 3% of \$50,000 = \$1,500. Employer adds \$1,500.

SIE Exam Tip: YOUR contributions are always yours. EMPLOYER match may have vesting.

Vesting Schedules

- Immediate vesting: 100% yours from day one
- Cliff vesting: 0% until X years, then 100%
- Graded vesting: Gradually increases (e.g., 20% per year)
- Applies to employer match only; leave before vested = forfeit

Example: 3-year cliff vesting. Leave at year 2 = forfeit all employer match.

SIE Exam Tip: Cliff = all or nothing. Graded = gradual.

401(k) Loans

- Borrow from your own account (if plan allows)
- Limit: Lesser of 50% of vested balance or \$50,000
- Must repay with interest (paid back to yourself)
- Leave job = loan typically due in full
- If not repaid = treated as distribution (tax + penalty)

SIE Exam Tip: 401(k) loans are from YOUR money. Not repaid = taxable distribution.

Hardship Withdrawals

- Withdraw for immediate financial need
- Must demonstrate hardship (medical, eviction, funeral, etc.)
- Subject to income tax
- May be subject to 10% penalty
- Reduces retirement savings permanently

SIE Exam Tip: Hardship = last resort. Still taxed, possibly penalized.

Rollovers When Leaving Employer

- Direct rollover to new 401(k): No tax
- Direct rollover to IRA: No tax
- Indirect rollover: 20% withheld; 60-day rule applies
- Cash out: Income tax + 10% penalty if under 59½
- Leave in old plan: No tax (if plan allows)

SIE Exam Tip: Direct rollover = best option. Cash out = worst (taxes + penalty).

RMDs in Employer Plans

- RMDs begin at age 73 (same as Traditional IRA)
- Still-working exception: May delay RMDs if still employed at that company
- Exception does NOT apply if you own >5% of the company
- Roth 401(k): RMDs required — unlike Roth IRA
- Can roll Roth 401(k) to Roth IRA to avoid RMDs

SIE Exam Tip: Roth 401(k) HAS RMDs. Roth IRA does NOT. Important distinction.

Key Takeaways

- 401(k) = private employers; 403(b) = nonprofits; 457 = government
- 457 government plans have no early withdrawal penalty
- Employer match may have vesting requirements
- Direct rollover avoids taxes; cash out = worst option
- RMDs at 73 (exception: still working at that employer)
- Roth 401(k) has RMDs; Roth IRA does not

Annuities: Fixed & Variable

Learning Objectives:

- Distinguish fixed from variable annuities
- Understand accumulation vs. annuitization phases
- Know annuity taxation and surrender charges

What Is an Annuity?

- Insurance contract providing income (often for retirement)
- Issued by insurance companies
- Can provide income for life (longevity protection)
- Tax-deferred growth (non-qualified annuities)
- Two phases: Accumulation and Annuitization

SIE Exam Tip: Annuity = insurance product. Not SIPC protected (insurance, not brokerage).

Annuity Phases

Phase	What Happens
Accumulation	Money grows tax-deferred; no payments yet
Annuitization	Convert to income stream; begin receiving payments

- Can choose to annuitize or take lump sum
- Once annuitized, generally irrevocable

SIE Exam Tip: Accumulation = saving. Annuitization = receiving income.

Fixed Annuities

- Guaranteed interest rate (set by insurer)
- Principal protected
- Predictable, stable income
- Insurance company bears investment risk
- Best for: Conservative investors wanting guaranteed income

SIE Exam Tip: Fixed annuity = guaranteed rate. Insurer takes the risk. NOT a security.

Variable Annuities

- Returns based on underlying investments (subaccounts)
- Subaccounts similar to mutual funds
- No guaranteed return (can lose money)
- Investor bears investment risk
- Best for: Investors willing to accept market risk for growth potential

SIE Exam Tip: Variable annuity = investment risk on YOU. IS a security (must be registered).

Fixed vs. Variable Comparison

Feature	Fixed Annuity	Variable Annuity
Returns	Guaranteed rate	Based on investments
Risk bearer	Insurance company	Investor
Is it a security?	No	Yes
Registration	Insurance license only	Securities + insurance license
Potential	Lower, stable	Higher, volatile

SIE Exam Tip: Variable = security (prospectus required). Fixed = NOT a security.

Annuity Payout Options

Option	Description
Life only	Payments for your lifetime; nothing to beneficiaries
Life with period certain	Lifetime payments; minimum guaranteed period
Joint and survivor	Payments continue for surviving spouse
Period certain	Payments for fixed period (e.g., 20 years)

SIE Exam Tip: Life only = highest payment but nothing left at death.

Annuity Taxation

- Growth: Tax-deferred during accumulation
- Withdrawals: Earnings taxed as ordinary income (not capital gains)
- LIFO treatment: Earnings come out first (taxed), then principal
- 10% penalty: If withdrawn before age 59½
- No contribution limits (unlike IRAs)

SIE Exam Tip: Annuity earnings = ordinary income tax. Early withdrawal = 10% penalty.

Surrender Charges

- Back-end fee if you withdraw early
- Typically decreases over time (e.g., 7% year 1, 6% year 2...)
- Surrender period: Usually 5-10 years
- Separate from IRS 10% penalty
- Check schedule before buying

Example: 7-year surrender schedule: 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0%

SIE Exam Tip: Surrender charge = insurance company fee. 10% penalty = IRS fee. Both may apply.

Annuity Death Benefits

- Standard death benefit: Beneficiary receives account value
- Stepped-up death benefit: Locks in gains periodically
- Avoids probate (beneficiary designation)
- Earnings still taxable to beneficiary as ordinary income

SIE Exam Tip: Annuity death benefit passes to beneficiary—earnings are taxable.

Annuity Riders (Optional Features)

- Guaranteed Minimum Income Benefit (GMIB): Guarantees minimum income
- Guaranteed Minimum Withdrawal Benefit (GMWB): Guarantees withdrawal amount
- Long-term care rider: Provides LTC benefits
- Riders cost extra (reduce returns)

SIE Exam Tip: Riders add guarantees but increase costs.

Annuity Suitability

Suitable For

Not Suitable For

Long time horizon	Short-term needs
Tax-deferred growth needed	Already in tax-deferred account (IRA)
Guaranteed income desired	Need liquidity
Longevity protection	Young investors with long horizon

SIE Exam Tip: Annuity in an IRA = no additional tax benefit. Often unsuitable.

Key Takeaways

- Fixed annuity = guaranteed return, NOT a security
- Variable annuity = investment risk, IS a security
- Two phases: accumulation (growth) and annuitization (income)
- Earnings taxed as ordinary income (LIFO)
- Early withdrawal = 10% penalty + possible surrender charge

Education & Special Savings Accounts

Learning Objectives:

- Explain 529 plans and their tax advantages
- Understand Coverdell ESAs and their limitations
- Know ABLE accounts and custodial account basics

Tax-Advantaged Education Savings

- Several options to save for education expenses
- Tax-free growth when used for qualified expenses
- Different contribution limits and rules per account type
- Understanding the options helps with suitability questions

SIE Exam Tip: Education accounts = tax-free for qualified education expenses

529 Plans Overview

- State-sponsored education savings plans
- Two types: savings plans and prepaid tuition plans
- After-tax contributions; tax-free growth and withdrawals (qualified expenses)
- High contribution limits (varies by state, often \$300,000+)
- No income limits for contributors

SIE Exam Tip: 529 = state-sponsored, high limits, tax-free for education

529 Plan Features

Feature	Details
Contributions	After-tax (not federally deductible)
Growth	Tax-free
Withdrawals	Tax-free for qualified education expenses
Qualified expenses	Tuition, fees, books, room & board, K-12 (\$10,000/year)
Non-qualified withdrawal	Earnings taxed + 10% penalty

SIE Exam Tip: Non-qualified 529 withdrawal = tax on earnings + 10% penalty

529 Plan Rules

- Owner controls the account, not the beneficiary
- Beneficiary can be changed to another family member
- No age limit for the beneficiary
- Limited Roth IRA rollover allowed after 15 years (new rules)
- Gift tax: up to 5 years of gifts in one year (\$90,000 in 2024)

SIE Exam Tip: Owner controls 529 and can change beneficiary, making it flexible

Prepaid Tuition vs. Savings Plans

Type	How It Works	Risk
Prepaid Tuition	Lock in today's tuition rates	Limited to participating schools
Savings Plan	Invest in mutual fund-like options	Investment risk; no guarantee

- Savings plans are more common and flexible
- Prepaid plans may not cover all education expenses

SIE Exam Tip: Prepaid = lock in rates. Savings = investment growth potential

Coverdell ESA

- Education Savings Account (formerly "Education IRA")
- Contribution limit: \$2,000 per year per beneficiary
- Income limits apply to the contributor
- Contributions stop at age 18; funds must be used by age 30
- Covers both K-12 and higher education expenses

SIE Exam Tip: Coverdell = low limit (\$2,000) with age restrictions; 529 = higher limits, no age limit

529 vs. Coverdell Comparison

Feature	529 Plan	Coverdell ESA
Contribution limit	High (state-specific)	\$2,000/year
Income limits	None	Yes
Age limits	None	Must use by 30
Investment options	Limited	Broad
State tax deduction	Often yes	No

SIE Exam Tip: 529 = higher limits, more flexible. Coverdell = more investment options but restrictive

ABLE Accounts

- Achieving a Better Life Experience accounts
- For individuals with disabilities (onset before age 26)
- Tax-free growth and withdrawals for qualified disability expenses
- Contribution limit: \$18,000 per year (2024)
- Does not affect government benefits eligibility (up to \$100,000)

SIE Exam Tip: ABLE = disability savings, tax-free for qualified disability expenses

UGMA / UTMA Accounts

- Uniform Gift / Transfer to Minors Act custodial accounts
- NOT tax-advantaged; no special tax treatment
- Irrevocable gift to the minor
- Child gains control at age of majority (18 or 21)
- Counts as the child's asset, which can affect financial aid

SIE Exam Tip: UGMA/UTMA = custodial, NOT tax-advantaged, irrevocable, child controls at majority

Custodial Account Kiddie Tax

- First ~\$1,300 of child's unearned income: tax-free
- Next ~\$1,300: taxed at the child's rate
- Above ~\$2,600: taxed at the parent's rate (kiddie tax)
- Applies until age 19, or age 24 if a full-time student

SIE Exam Tip: Kiddie tax = child's unearned income above the threshold taxed at parent's rate

Choosing the Right Account

- College savings, high contributions: 529 Plan
- K-12 plus college, more investment options: Coverdell ESA
- Disability-related expenses: ABLE Account
- General gift to a minor (non-education): UGMA / UTMA

SIE Exam Tip: Match the account type to the goal — classic suitability question pattern

Key Takeaways

- 529 plans: high limits, tax-free for education, state-sponsored
- Coverdell ESAs: \$2,000 limit, age restrictions, more investment options
- ABLE accounts: for disabilities, tax-free for qualified disability expenses
- UGMA / UTMA: custodial accounts, NOT tax-advantaged, irrevocable
- Non-qualified withdrawals = taxes plus 10% penalty on earnings

Order Types: How to Buy and Sell

Learning Objectives:

- Distinguish market, limit, stop, and stop-limit orders
- Understand what each order type guarantees (and doesn't)
- Explain order duration

Why Order Types Matter

- Different orders = different execution and risk
- Choosing the right order protects your interests
- Essential knowledge for any securities transaction

SIE Exam Tip: Know what each order guarantees—execution OR price, never both.

Market Orders

- Buy or sell immediately at best available price
- Guarantees: Execution
- Does NOT guarantee: Price
- Use when speed matters more than exact price

Example: Market order to buy 100 shares at \$50.00, filled at \$50.15 because the market moved.

SIE Exam Tip: Market orders guarantee execution, NOT price.

Limit Orders

- Set a maximum price (buy) or minimum price (sell)
- Guarantees: Price (or better)
- Does NOT guarantee: Execution
- Buy limit: Placed BELOW market → executes when price drops
- Sell limit: Placed ABOVE market → executes when price rises

SIE Exam Tip: Limit orders guarantee price, NOT execution. Opposite of market orders.

Stop Orders (Stop-Loss)

- Trigger order: Becomes a market order when stop price is reached
- Sell stop: Placed BELOW current price → limits losses
- Buy stop: Placed ABOVE current price → protects short positions
- Does NOT guarantee execution price after triggered

Example: Sell stop at \$45 triggers market order. You might get \$44.80.

SIE Exam Tip: Stop orders become MARKET orders when triggered—no price guarantee after.

Stop-Limit Orders

- Two prices: Stop price AND limit price
- When stop price reached, becomes a limit order (not market)
- Gives price protection after trigger
- Risk: May NOT execute if price moves past limit

Example: Stop \$45, Limit \$44. Won't sell below \$44. If stock gaps to \$43, no fill.

SIE Exam Tip: Stop-limit gives price protection but risks no execution if price gaps through.

Order Type Summary Table

Order Type	Guarantees	Risk	Placement
Market	Execution	Bad price	N/A
Buy Limit	Price	No execution	Below market
Sell Limit	Price	No execution	Above market
Sell Stop	Triggers action	Bad price after trigger	Below market
Buy Stop	Triggers action	Bad price after trigger	Above market
Stop-Limit	Price after trigger	No execution if gaps	Varies

SIE Exam Tip: Memorize this table—foundation of order type questions.

Order Duration

- Day Order: Expires at end of trading day if not filled (DEFAULT)
- Good-Til-Canceled (GTC): Remains open until filled or canceled
- Fill-or-Kill (FOK): Fill entire order immediately or cancel all
- All-or-None (AON): Fill entire order or none (can wait)

SIE Exam Tip: Day orders are default. GTC stays open. FOK = all now or nothing.

Key Takeaways

- Market order = execution guaranteed, price not
- Limit order = price guaranteed, execution not
- Stop orders trigger at price, then become market orders
- Stop-limit adds price protection but may not fill
- Day orders expire; GTC stays open

Market Structure: Where Trading Happens

Learning Objectives:

- Distinguish primary from secondary markets
- Explain exchanges, OTC markets, and ECNs
- Understand market makers, bid/ask, and the spread

Primary vs. Secondary Markets

- Primary Market: New securities issued (IPOs, new bonds)
- Issuer receives the proceeds from the sale
- Secondary Market: Existing securities traded between investors
- Seller receives the proceeds (not the issuer)

Example: Company IPOs at \$20/share = primary. You buy from another investor at \$25 = secondary

SIE Exam Tip: Primary = issuer gets money. Secondary = investors trade with each other

Types of Secondary Markets

- Exchanges: Centralized, regulated venues (NYSE, Nasdaq)
- OTC: Decentralized dealer network (less regulated)
- Third Market: Listed securities traded OTC (institutional)
- Fourth Market: Institution-to-institution (dark pools, no broker)

SIE Exam Tip: Know all four market types and what makes each different

Stock Exchanges

- NYSE (New York Stock Exchange)
- Auction market with Designated Market Makers (DMMs)
- Physical floor + electronic trading
- Nasdaq: Dealer market with multiple competing market makers
- Fully electronic—no physical trading floor

SIE Exam Tip: NYSE = auction/DMM. Nasdaq = dealer/multiple market makers

Over-the-Counter (OTC) Markets

- Decentralized: No central exchange floor
- Dealer network: Dealers quote prices and trade
- OTC Bulletin Board (OTCBB): Electronic quotation system
- Pink Sheets (OTC Pink): Less regulated, higher risk
- Typically smaller companies, less liquidity

SIE Exam Tip: OTC = dealer market. Less regulation than exchanges. Higher risk

Bid, Ask, and Spread

- Bid: Highest price a buyer will pay
- Ask (Offer): Lowest price a seller will accept
- Spread: Difference between bid and ask = dealer's profit
- You BUY at the ASK (higher) | You SELL at the BID (lower)

Example: Bid \$49.90, Ask \$50.10 — Spread = \$0.20

SIE Exam Tip: You BUY at the ASK, SELL at the BID. Spread is dealer's profit

What the Spread Tells You

- Narrow spread (\$0.01): High liquidity, active trading (blue-chip stocks)
- Wide spread (\$0.50+): Low liquidity, less trading (thinly traded stocks)
- Spread narrows with competition among dealers
- Wider spread = harder to trade at favorable prices

SIE Exam Tip: Wider spread = less liquid

Market Makers

- Dealers who stand ready to buy AND sell
- Quote two-sided markets (bid and ask)
- Profit from the spread
- Required to maintain continuous quotes
- Provide liquidity to the market

SIE Exam Tip: Market makers must quote both bid AND ask - can't quote just one side

ECNs (Electronic Communication Networks)

- Computerized systems matching buy/sell orders
- Alternative Trading Systems (ATS)
- Match orders directly without market maker
- Often faster execution
- Examples: Instinet, ARCA

SIE Exam Tip: ECNs match orders electronically—no dealer intermediary

Third & Fourth Markets and Dark Pools

- Third Market: Exchange-listed securities traded OTC
- Used by institutions and broker-dealers
- Fourth Market: Direct institution-to-institution trading
- Dark Pools: Private exchanges with no public quotes (block trades, anonymity)
- Fourth market and dark pools reduce market impact of large orders

SIE Exam Tip: Third = listed stocks traded OTC. Fourth/Dark pools = institution-to-institution

Market Structure - Key Takeaways

- Primary market = new issuance; Secondary = trading
- NYSE = auction/DMM; Nasdaq = dealer/market makers
- OTC = decentralized dealer network (higher risk)
- Bid = buy price; Ask = sell price; Spread = dealer profit
- ECNs match orders electronically
- Third market = listed OTC; Fourth = institution-to-institution

Trade Execution, Settlement & Short Selling

Learning Objectives:

- Explain how trades are executed and reported
- Understand the settlement cycle (T+1)
- Know short selling basics and risks

Trade Lifecycle Overview

Order Placed → Execution → Clearing → Settlement

Step	What Happens
Order	Customer places order
Execution	Order filled (buyer/seller matched)
Clearing	Trade details confirmed, obligations calculated
Settlement	Securities and money exchange hands

SIE Exam Tip: Know the sequence: Order → Execution → Clearing → Settlement

Trade Execution

- Execution = when the trade is completed
- Trade is reported to the tape (public record)
- Customer receives trade confirmation
- Execution price may differ from quoted price (especially market orders)

SIE Exam Tip: Execution is when the trade happens, not when it settles

Best Execution Obligation

- Broker-dealers must seek best execution for customers
- Consider: price, speed, likelihood of execution
- Must use "reasonable diligence"
- Cannot route orders just for their own benefit
- FINRA Rule 5310

SIE Exam Tip: Best execution = broker must try to get the best deal for the customer

What Is Settlement?

- Settlement = actual exchange of securities and payment
- Buyer receives securities; Seller receives payment
- Occurs AFTER execution (not same day)
- Ownership transfers on settlement date

SIE Exam Tip: Settlement is when ownership officially transfers and payment is made

Settlement Cycles

Security Type	Settlement
Stocks (equities)	T+1
Corporate bonds	T+1
Municipal bonds	T+1
Government securities	T+1
Options	T+1
Cash settlement	T+0 (same day, special request)

- T = Trade date; +1 = One business day after
- Trade Monday → Settle Tuesday
- Weekends and holidays don't count

SIE Exam Tip: T+1 is standard ("regular way") for most securities

Trade Confirmations

- Written notice of trade details sent to customer
- Must include: Security name/quantity, Price/total, Commission/fees
- Must include: Settlement date, Agent vs. principal capacity
- Must be sent at or before settlement

SIE Exam Tip: Confirmations must disclose if firm acted as agent (commission) or principal (markup)

Clearing Organizations

Organization	Role
NSCC (National Securities Clearing Corporation)	Clears equities
DTCC (Depository Trust & Clearing Corporation)	Parent company
OCC (Options Clearing Corporation)	Clears options

- Clearinghouses **guarantee** trades will settle
- Reduce counterparty risk for all parties

SIE Exam Tip: Clearinghouses guarantee settlement—reduce risk

What Is Short Selling?

- Selling securities you don't own
- Borrow shares → Sell → Buy back later → Return shares
- Profit if price falls
- Bearish strategy

Example: Borrow and sell 100 shares at \$50. Price drops to \$40. Buy back at \$40. Profit = \$10/share

SIE Exam Tip: Short selling = sell first, buy later. Profit when price falls

Short Selling Requirements & Risks

- Must be in margin account (cannot short in cash account)
- Must locate shares before shorting (Reg SHO)
- 50% initial margin (Regulation T)
- UNLIMITED loss potential (price can rise infinitely)
- Dividend liability (pay dividends to lender); May be forced to cover

SIE Exam Tip: Short selling has UNLIMITED risk and requires margin account

Circuit Breakers

- Automatic halts when market drops significantly
- Based on S&P 500 decline from prior close

Level	Decline	Action
Level 1	7%	15-minute halt
Level 2	13%	15-minute halt
Level 3	20%	Trading halted for day

SIE Exam Tip: Circuit breakers: 7%, 13%, 20% drops trigger halts

Trading Halts

- Temporary pause in trading a specific security
- Reasons: Pending news announcement
- Reasons: Order imbalance, Regulatory concerns
- Exchange or SEC can halt trading
- Protects investors from trading on incomplete information

SIE Exam Tip: Halts protect investors when significant news is pending

Trade Execution - Key Takeaways

- Trade lifecycle: Order → Execution → Clearing → Settlement
- T+1 is regular way settlement for most securities
- Brokers owe best execution to customers
- Confirmations must disclose agent vs. principal capacity
- Short selling = unlimited risk, requires margin account, must locate shares
- Circuit breakers halt market at 7%, 13%, 20% drops

Regulatory Framework & SROs

Learning Objectives:

- Identify the major regulators and their jurisdictions
- Distinguish federal regulators from SROs
- Understand registration, qualification, and CE requirements

The Regulatory Hierarchy

- Federal Government: SEC at the top
- SROs operate under SEC supervision
- State Regulators: Parallel authority through Blue Sky laws
- Each level has distinct jurisdiction and powers

SIE Exam Tip: SEC = government. FINRA, MSRB, CBOE, SIPC = NOT government agencies

The SEC — Federal Oversight

- Created by Securities Exchange Act of 1934
- Federal government agency (NOT an SRO)
- Mission: Protect investors, maintain fair markets, facilitate capital formation
- Oversees all SROs including FINRA and MSRB

SIE Exam Tip: SEC has CIVIL authority only. Criminal cases go to DOJ

FINRA — Regulating Broker-Dealers

- Largest SRO for the securities industry
- Regulates broker-dealers and registered representatives
- Administers qualification exams (including the SIE)
- Can fine, suspend, or bar industry professionals

SIE Exam Tip: FINRA is the primary SRO. It writes AND enforces its rules

MSRB — Municipal Securities

- Writes rules for municipal securities dealers and advisors
- Operates EMMA (Electronic Municipal Market Access)
- KEY EXAM POINT: MSRB CANNOT enforce its own rules
- FINRA and SEC enforce MSRB rules for broker-dealers

SIE Exam Tip: MSRB writes rules but does NOT enforce them

Other Regulatory Bodies

- CBOE: Options exchange regulation (SRO)
- Federal Reserve Board: Sets margin (Reg T); monetary policy
- State Regulators: Blue Sky laws; in-state offerings
- NASAA: Coordinates state regulators (NOT a regulator)
- Department of Justice: Criminal prosecution of securities fraud

SIE Exam Tip: NASAA coordinates state regulators but is NOT a regulator itself

SIPC — Customer Protection

- Created by SIPA of 1970; protects customers if a broker-dealer fails
- NOT a government agency; NOT an SRO
- Coverage: \$500,000 total per customer / \$250,000 cash limit
- Does NOT protect against market losses

SIE Exam Tip: SIPC = firm failure protection only. Distinct from FDIC

Registration of Associated Persons

- Registered persons: Engage in transactions, make recommendations
- Form U4: Application for registration
- Form U5: Termination notice
- Background checks and fingerprinting required

SIE Exam Tip: Form U4 = hire; Form U5 = termination. Both filed with FINRA

Qualification Exams

- SIE: Securities Industry Essentials (entry-level, no sponsorship needed)
- Series 6: Investment company / variable products limited rep
- Series 7: General securities representative
- Series 63 / 65 / 66: State law and investment adviser exams

SIE Exam Tip: SIE alone does NOT qualify you to sell securities. Need a top-off

Continuing Education

- Regulatory Element: Industry-wide rules training
- Required annually for all registered persons (2023+)
- Firm Element: Firm-specific training program
- Covers products, services, and rules relevant to the job

SIE Exam Tip: Regulatory = industry rules; Firm = firm-specific

Reportable Events on Form U4

- Criminal charges/convictions (felonies, certain misdemeanors)
- Regulatory actions (SEC, FINRA, state) and customer complaints
- Bankruptcy (within 10 years), liens, and judgments
- Outside Business Activities (OBAs)

SIE Exam Tip: Updates required promptly (typically within 30 days)

Regulatory Framework - Key Takeaways

- SEC = federal government agency overseeing capital markets
- FINRA = largest SRO; writes AND enforces broker-dealer rules
- MSRB writes municipal rules but does NOT enforce them
- SIPC protects against firm failure (\$500K / \$250K cash)
- Form U4 = registration; Form U5 = termination; CE has 2 elements

Customer Accounts & Account Types

Learning Objectives:

- Identify types of customer accounts and their features
- Understand account opening requirements and KYC
- Distinguish discretionary from non-discretionary accounts
- Know margin account basics

Account Opening Requirements

- New Account Form captures customer information
 - Required: name, address, DOB, SSN/TIN, employment, financial info, objectives
- Customer signature is NOT required to open the account
- Principal (supervisor) approval IS required
- USA PATRIOT Act mandates a Customer Identification Program (CIP)

SIE Exam Tip: Customer signature not required to open. Principal approval IS required.

Customer Identification Program

- Required by USA PATRIOT Act, Section 326
- Identity must be verified BEFORE opening the account
- Four required pieces of information:
 - Name, date of birth, physical address, ID number (SSN/TIN/passport)
- Customer screened against OFAC government watch lists

SIE Exam Tip: CIP requires 4 pieces of info — verified before opening.

Cash vs. Margin Accounts

Feature	Cash Account	Margin Account
Payment	Full payment required	Can borrow from firm
Settlement	Full payment by T+1	Initial deposit + loan
Short Selling	Not permitted	Required for shorts
Reg T	N/A	50% initial margin
Hypothecation	No	Yes (collateral)

SIE Exam Tip: Short sales REQUIRE a margin account.

Account Registration Types

Type	Description
Individual	One owner
Joint Tenants (JTWROS)	Equal ownership; survivor inherits
Tenants in Common (TIC)	Specified % ownership; passes to estate
Custodial (UGMA/UTMA)	Adult manages for minor
Trust	Trustee manages for beneficiaries
Corporate	Owned by corporation
Partnership	Owned by partnership

SIE Exam Tip: JTWROS = survivorship. TIC = passes to estate.

Discretionary vs. Non-Discretionary

Discretionary

- Rep can trade without prior customer approval
- Requires WRITTEN authorization from customer
- Requires firm/principal approval
- Each trade marked "discretionary"
- Subject to heightened supervision

Non-Discretionary

- Customer approves each trade before execution
- Rep gives advice; customer decides
- Default for most accounts

SIE Exam Tip: Discretion needs WRITTEN authorization.

Time and Price Discretion

- “Buy 100 XYZ at a good price today” — NOT full discretion
- Rep choosing only WHEN and at WHAT price is permitted
- Customer must specify: action, security, and quantity
- Authority is good for that day only
- Does NOT require written authorization

SIE Exam Tip: Time/price discretion ≠ discretionary account.

Power of Attorney (POA)

Type	Authority	Survives Incapacity
Limited POA	Specific actions only	Depends on document
Full POA	All account actions	Depends on document
Durable POA	Continues if grantor incapacitated	YES
Non-Durable POA	Ends if grantor incapacitated	NO

- ALL POAs terminate at death — no exceptions
- Trading authorization is a form of limited POA

SIE Exam Tip: All POAs end at death. Durable survives incapacity.

Trusted Contact Person

- FINRA Rule 4512: firms must REQUEST a trusted contact
- Person aged 18+ that the firm can contact about the account
- Customer is NOT required to provide one — only required to be asked
- Firm may disclose information about suspected exploitation
- Especially valuable for senior investors
- Trusted contact is NOT authorized to trade

SIE Exam Tip: Trusted contact $\neq$ POA. No trading authority.

Margin Account Basics

- Regulation T — Federal Reserve rule governing securities credit
- 50% initial margin requirement on equity purchases
- Customer borrows the rest from broker; securities are collateral
- Required documents:
 - Margin Agreement (must sign), Hypothecation, Credit Agreement
 - Loan Consent — optional

SIE Exam Tip: Reg T = 50% initial margin. Margin Agreement required.

Account Statements & Confirmations

- Trade confirmation: details of each trade — at or before settlement
- Confirmations disclose agent vs. principal capacity of the firm
- Account statements: monthly when there is account activity
- Statements: quarterly minimum even with no activity
- Statements show positions, transactions, and cash balance

SIE Exam Tip: Statements quarterly min., monthly with activity.

Senior Investor Protection

- FINRA Rule 2165 — Financial Exploitation of Specified Adults
- Applies to investors aged 65 or older, OR
- Adults 18+ unable to protect interests due to impairment
- Permits temporary HOLD on disbursements — up to 15 business days
- Hold may be extended after internal review
- Trusted contact must be notified when hold is placed

SIE Exam Tip: Rule 2165 allows TEMPORARY hold to investigate.

Customer Accounts — Key Takeaways

- CIP verifies name, DOB, address, ID number BEFORE opening
- JTWROS = survivorship; TIC = passes to estate
- Discretion requires WRITTEN authorization
- All POAs terminate at death; durable survives incapacity
- Trusted contact = informational only — no trading authority
- Reg T = 50% initial margin; Margin Agreement required
- Rule 2165 protects seniors with temporary hold authority

Suitability, Reg BI & Customer Communications

Learning Objectives:

- Understand the suitability obligation and Reg BI
- Apply Know Your Customer (KYC) requirements
- Distinguish types of customer communications
- Recognize advertising and telemarketing rules

Know Your Customer – Rule 2090

- Firm must use reasonable diligence to know the customer
- Know essential facts about the customer and their accounts
- Verify authority of anyone acting on customer's behalf
- Required at account opening AND on an ongoing basis

SIE Exam Tip: KYC = Rule 2090. Suitability = Rule 2111. Different but related

Suitability – FINRA Rule 2111

- Reasonable Basis: recommendation suitable for SOME investors
- Customer-Specific: suitable for THIS particular customer
- Quantitative: series of trades not excessive (anti-churning)
- Applies to recommendations, not unsolicited orders

Three obligations: Reasonable Basis, Customer-Specific, Quantitative

SIE Exam Tip: Quantitative suitability addresses churning

Customer Investment Profile

- Age, financial situation, tax status
- Investment objectives, experience, time horizon
- Liquidity needs and risk tolerance
- Other investments held outside the firm

SIE Exam Tip: The investment profile drives every suitability decision

Regulation Best Interest (Reg BI)

- SEC rule effective June 2020
- Applies to broker-dealers making recommendations to RETAIL customers
- Must act in customer's BEST INTEREST – cannot put firm first
- Four obligations: Disclosure, Care, Conflict of Interest, Compliance

SIE Exam Tip: Reg BI applies to RETAIL customers. Higher than suitability but not fiduciary

Reg BI vs. Fiduciary Duty

Standard	Applies To	Requirement
Suitability	Pre-Reg BI BD	Suitable for customer
Reg BI	Broker-dealers	Customer's best interest
Fiduciary	Investment advisers	Highest standard; loyalty + care

- Investment advisers (RIAs) owe fiduciary duty under Advisers Act of 1940
- Broker-dealers owe Reg BI standard
- Both higher than old suitability rule

SIE Exam Tip: BD = Reg BI. IA = Fiduciary. Don't confuse them.

Form CRS

- Required by Reg BI – plain English disclosure document
- Provided to retail investors at or before recommendation
- Covers: relationship, fees, conflicts, disciplinary history
- Discloses standard of conduct (Reg BI for BDs)

SIE Exam Tip: Form CRS = Reg BI disclosure. Provided to retail investors

Types of Customer Communications

Type	Definition	Approval
Retail Communication	To 25+ retail investors in 30 days	Principal pre-approval
Correspondence	To 25 or fewer retail investors	Post-use review/sample
Institutional	To institutional investors only	Supervisory procedures

- Retail communications include: ads, websites, social media, sales literature
- Different rules apply to each category

SIE Exam Tip: 25 retail investors in 30 days = the dividing line

Advertising Standards – Rule 2210

- Must be fair, balanced, and not misleading
- Cannot guarantee performance – past performance not predictive
- Material claims must be substantiated
- Risk disclosures required where appropriate

SIE Exam Tip: No guarantees of performance. Past performance $\neq$ future results

Telemarketing – Rule 3230

- Calling hours: 8:00 AM to 9:00 PM customer's local time
- Must check National DNC Registry and firm's DNC list
- Must identify caller, firm, and purpose of call
- Exception: existing customer (within 18 months of last transaction)

SIE Exam Tip: 8 AM–9 PM customer's local time. 18-month existing customer exception

Social Media Communications

- Static content (profile, bio): treated as advertising
- Interactive content (posts, replies): treated as correspondence
- Firms must retain records of business communications
- Personal devices used for business are subject to supervision

SIE Exam Tip: Static = advertising; Interactive = correspondence

Suitability & Reg BI - Key Takeaways

- KYC = Rule 2090; Suitability = Rule 2111; Reg BI = SEC rule
- Three suitability obligations: Reasonable Basis, Customer-Specific, Quantitative
- Reg BI applies to BDs and retail customers; IAs owe fiduciary duty
- Form CRS = required disclosure to retail investors
- 25 retail investors / 30 days threshold; calling hours 8 AM–9 PM local

Prohibited Practices & Ethical Conduct

Learning Objectives:

- Identify prohibited trading and sales practices
- Understand insider trading rules and penalties
- Recognize OBA and private securities transaction rules
- Know rules on gifts and political contributions

Why Prohibited Practices Matter

- FINRA regulates rep conduct to protect investors
- Some actions are unethical even if not criminal
- Violations can lead to fines, suspension, bars, or criminal prosecution
- Rep is responsible — "my manager didn't stop me" is NOT a defense

SIE Exam Tip: Both ethics and rules matter — know the purpose, not just the rule

Insider Trading

- Trading on Material, Non-Public Information (MNPI)
- MATERIAL = would affect a reasonable investor's decision
- NON-PUBLIC = not disseminated to the general public
- Illegal under Exchange Act of 1934, Rule 10b-5; both tipper and tippee liable

Example: CEO learns of a pending merger, tells a friend who buys stock — both liable

SIE Exam Tip: Insider trading = MNPI. Both tipper and tippee are liable

Insider Trading: Penalties

- Civil: up to 3x the profit gained or loss avoided
- Criminal: up to \$5M fine + 20 years prison (individual); \$25M for entities
- Disgorgement of profits and bar from the securities industry
- Section 20A creates contemporaneous trader liability

SIE Exam Tip: Civil = 3x profit. Criminal = up to 20 years

Manipulation & Fraud

Practice	Description
Front-running	Trading ahead of customer order or research
Painting the tape	Fake trades to create appearance of activity
Marking the close	Trades to affect closing price
Pump and dump	Inflate price then sell
Matched orders	Coordinated buy/sell to mimic activity
Spoofing	Place orders intending to cancel

SIE Exam Tip: Front-running and painting the tape are the most common manipulation types tested

Churning

- Excessive trading in a customer account to generate commissions
- Violates quantitative suitability under FINRA Rule 2111
- Indicators: high turnover, frequent in-and-out trades, oversized commissions

SIE Exam Tip: Churning = excessive trading for commissions.
Discretionary accounts at higher risk — still prohibited

Unauthorized Trading

- Placing trades without the customer's prior approval
- Allowed only with written discretionary authorization or same-day time/price discretion
- Even profitable trades can be unauthorized
- Always a violation regardless of outcome

SIE Exam Tip: Unauthorized trading is a violation EVEN IF the trade is profitable

Outside Business Activities (OBAs)

- FINRA Rule 3270: any work outside the firm requires written notice
- Includes both paid and unpaid activities
- Even non-securities work counts (e.g., real estate agent, consultant)
- Firm must review and may restrict; rep must update if activities change

SIE Exam Tip: OBA = any outside work. Written notice required, paid or not

Private Securities Transactions (PSTs)

- FINRA Rule 3280: securities transactions outside the firm ("selling away")
- WITH compensation: requires firm approval; firm must supervise and record
- WITHOUT compensation: requires written notice; firm may restrict
- Key distinction: OBA = non-securities work; PST = securities away from firm

SIE Exam Tip: OBA $\neq$ PST. Compensation makes a PST higher-risk and more strictly supervised

Gifts, Gratuities & Entertainment

- FINRA Rule 3220: \$300 per person, per year gift limit
- Applies to gifts to or from customers and business contacts
- Reasonable business entertainment (meals, events) excluded from the limit
- Bribery is judged by intent — anything meant to influence is prohibited

SIE Exam Tip: \$300 annual gift limit. Entertainment ≠ gift if reasonable

Political Contributions (Pay-to-Play)

- MSRB Rule G-37 (municipal); SEC Rule 206(4)-5 (investment advisers)
- Limits contributions to officials who can award municipal business
- \$250 limit if rep can vote for the official; \$350 if rep cannot
- Violation = 2-year ban on muni securities business with that issuer

SIE Exam Tip: G-37 = pay-to-play. \$250/\$350 limits, 2-year business ban

Borrowing & Lending With Customers

- FINRA Rule 3240: borrowing/lending between rep and customer is generally prohibited
- Limited exceptions: immediate family members
- Certain financial institution and personal/business relationships also exempted
- Even when permitted, firm approval and written disclosure are required

SIE Exam Tip: Borrowing/lending with customers usually prohibited; family is the main exception

Prohibited Practices - Key Takeaways

- Insider trading = MNPI; tipper and tippee both liable; 3x civil penalty
- Front-running, churning, painting the tape are prohibited manipulations
- Unauthorized trading is a violation even if profitable
- OBA = outside work (notice); PST = outside securities (approval if paid)
- \$300 annual gift limit; G-37 political: \$250/\$350; borrowing/lending prohibited

AML, Privacy & Customer Protection

Learning Objectives:

- Understand AML requirements and reporting thresholds
- Recognize the three stages of money laundering
- Know privacy rules under Regulation S-P
- Identify books and records requirements

AML Compliance Overview

- Bank Secrecy Act (BSA) + USA PATRIOT Act
- FINRA Rule 3310: Firms must have AML compliance program

Required components:

1. Written policies and procedures
2. Designated AML compliance officer
3. Annual independent testing
4. Ongoing employee training
5. Customer Identification Program (CIP)

SIE Exam Tip: Rule 3310 = AML program. Includes Customer Identification Program (CIP)

Three Stages of Money Laundering

Stage	What Happens	Example
Placement	Introducing illegal funds into system	Cash deposits
Layering	Concealing source through transactions	Wire transfers, trades
Integration	Using laundered money for legit use	Buy real estate

- Each stage gets harder to detect;
- AML programs target all three

SIE Exam Tip: Memorize the order: Placement → Layering → Integration

Currency Transaction Reports (CTR)

- Filed for cash transactions over \$10,000
- Single transaction OR aggregated same-day transactions
- Filed with FinCEN (Financial Crimes Enforcement Network)
- NOT filed for normal securities transactions—only cash movements

SIE Exam Tip: CTR threshold = \$10,000 cash. Filed with FinCEN

Suspicious Activity Reports (SAR)

- Filed for suspicious transactions of \$5,000 or more
- Must file within 30 calendar days of detection (with FinCEN)
- CONFIDENTIAL: Cannot tell customer SAR was filed (no "tipping off")
- Indicators: structuring (just under \$10K), unusual activity, possible terrorist financing

SIE Exam Tip: SAR threshold = \$5,000. Cannot tip off the customer. 30-day filing deadline

OFAC & Sanctions

- Office of Foreign Assets Control (Treasury Dept.) administers economic and trade sanctions
- Maintains the SDN List (Specially Designated Nationals); firms must screen customers against it
- Cannot do business with sanctioned individuals/entities; violations bring severe penalties

SIE Exam Tip: OFAC = sanctions. Screen customers against SDN list at account opening

Regulation S-P (Privacy)

- SEC rule protecting customer non-public personal information
- Privacy notice required at account opening
- Annual privacy notice (unless exempt)
- Opt-out option for sharing with non-affiliates; safeguard customer info

SIE Exam Tip: Reg S-P = privacy. Opt-out for non-affiliate sharing. Annual notice required

Identity Theft Red Flags (Reg S-ID)

- Firms must develop an Identity Theft Prevention Program
- Identify, detect, prevent, and respond to red flags
- Red flag examples: suspicious documents, address discrepancies
- Also: unusual account activity, customer notifications of fraud

SIE Exam Tip: Reg S-ID requires firms to detect and prevent identity theft

Books & Records Requirements

- SEC Rule 17a-3: WHAT records must be created
- SEC Rule 17a-4: HOW LONG records must be retained

Record Type	Retention
Customer account records	6 years (last 2 easily accessible)
Trade blotters	6 years
Order tickets	3 years
Communications	3 years
Forms U4 / U5	3 years after termination

SIE Exam Tip: 17a-3 = creation; 17a-4 = retention. 6 years for customer records

Customer Complaints

- FINRA Rule 4513: written customer complaints must be retained
- Records kept at OSJ (Office of Supervisory Jurisdiction)
- Required: complainant ID, date received, nature, disposition
- Retain for 4 years
- Pattern of complaints = red flag for misconduct

•SIE Exam Tip: Written complaints must be retained for 4 years per Rule 4513.

AML & Privacy - Key Takeaways

- AML program: policies, officer, testing, training, CIP (Rule 3310)
- Three stages: Placement → Layering → Integration
- CTR = \$10K cash; SAR = \$5K suspicious; both filed with FinCEN
- Cannot tip off customer about SAR; OFAC screens against SDN list
- 17a-3 (creation) and 17a-4 (retention); customer records 6 years